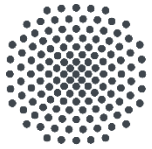
CMOS Analog Integrated Circuit Design

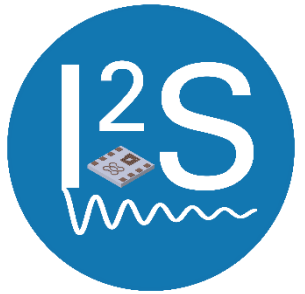
chapter04_Single Stage Amplifiers

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University of Stuttgart

Winter Term 2025/26



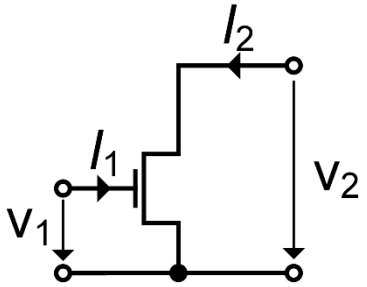
University of Stuttgart
Institute of Smart Sensors



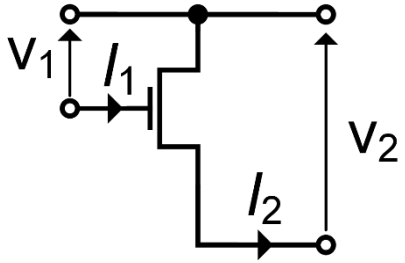
Overview

- **Common-Source (CS) Amplifier**
  - **Amplification – Quick Review / CS Amplifier with Resistor Load**
  - CS Amplifier with Diode-Connected Transistor Load
  - CS Amplifier with Current Source Load
  - CS Amplifier with Cascode Current Source Load
  - Full Cascode CS Amplifier
- Common-Drain (CD) Amplifier
- Common-Gate (CG) Amplifier

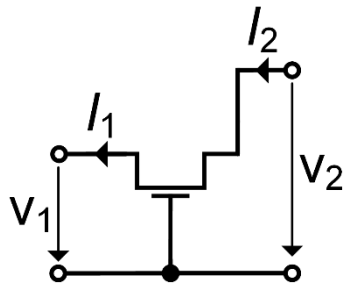
MOSFET – Configurations



common-source configuration (CS)



common-drain configuration (CD)

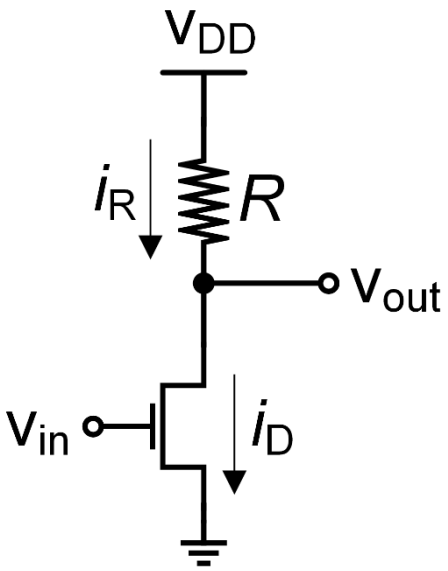


common-gate configuration (CG)

	common-source	common-drain	common-gate
voltage amplification	high	≈ 1	high
current amplification	very high (→ ∞)	very high (→ ∞)	≈ 1
input resistance	very high (→ ∞)	very high (→ ∞)	low
output resistance	average	low	average
phase shift	-180°	0°	0°

Bias Point & Graphical Illustration of Amplification

R=100k, $V_{DD} = 1.2V$, W/L=10/10



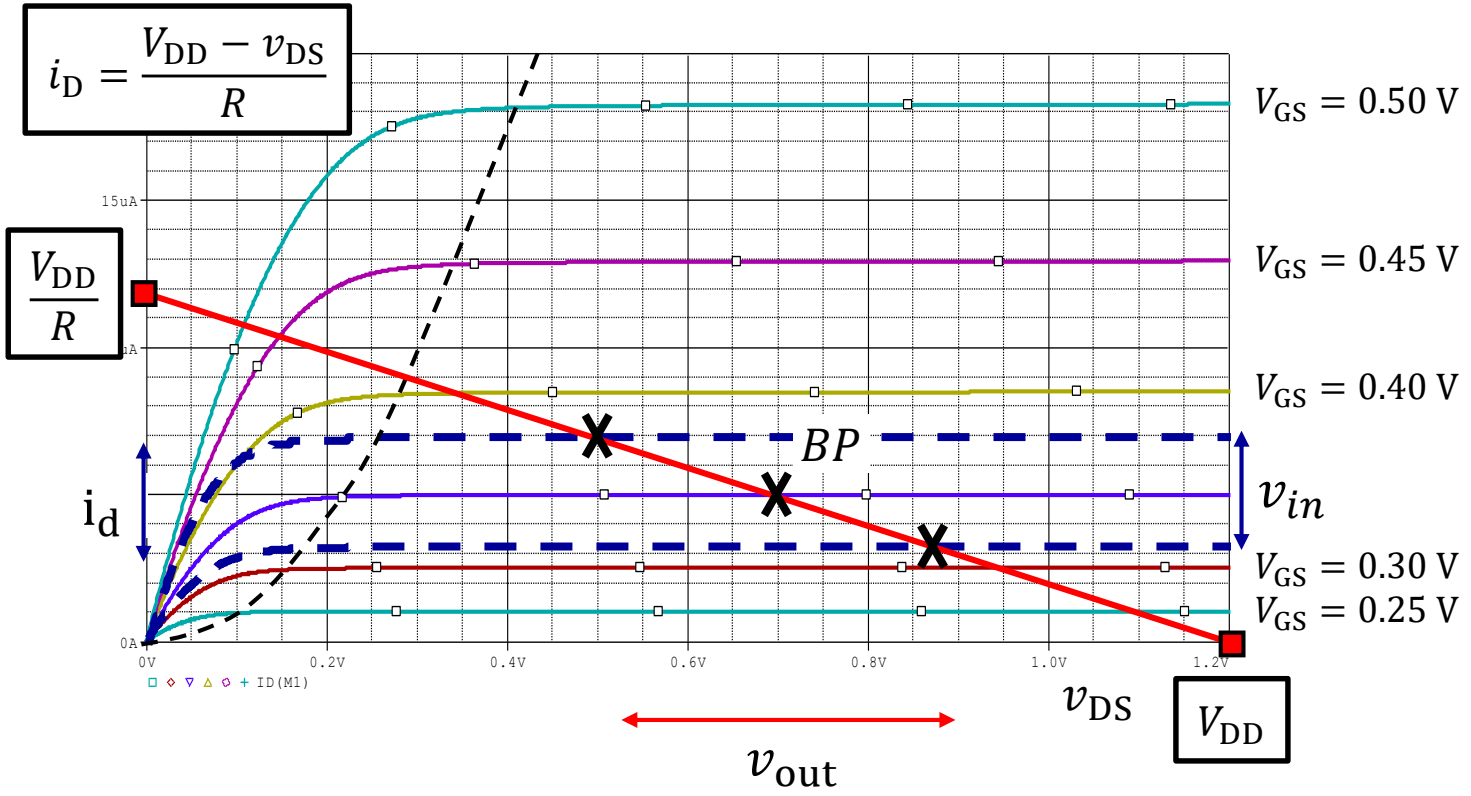
$$i_D = \frac{\beta_n}{2}(v_{IN} - V_{th})^2$$

$$v_{OUT} = V_{DD} - Ri_D$$

$$i_D = i_R$$

$$i_d = G_m v_{in}$$

$$v_{out} = -Ri_d = -G_m R v_{in}$$



$$v_{in} = v_{gs} \quad v_{in} \longrightarrow i_d$$
$$v_{out} = v_{ds} \quad i_d \longrightarrow v_{out}$$

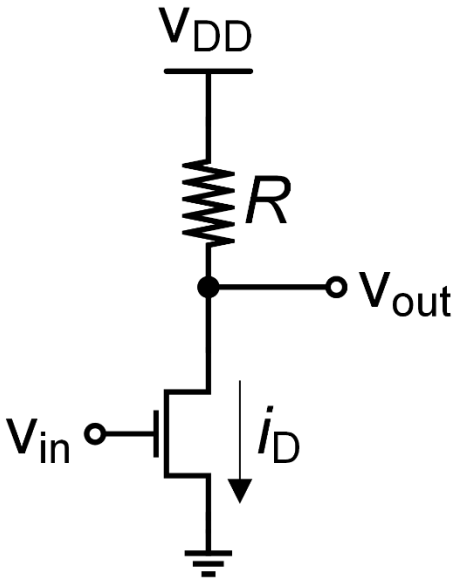
$$A_V = \frac{v_{out}}{v_{in}}$$

transconductance

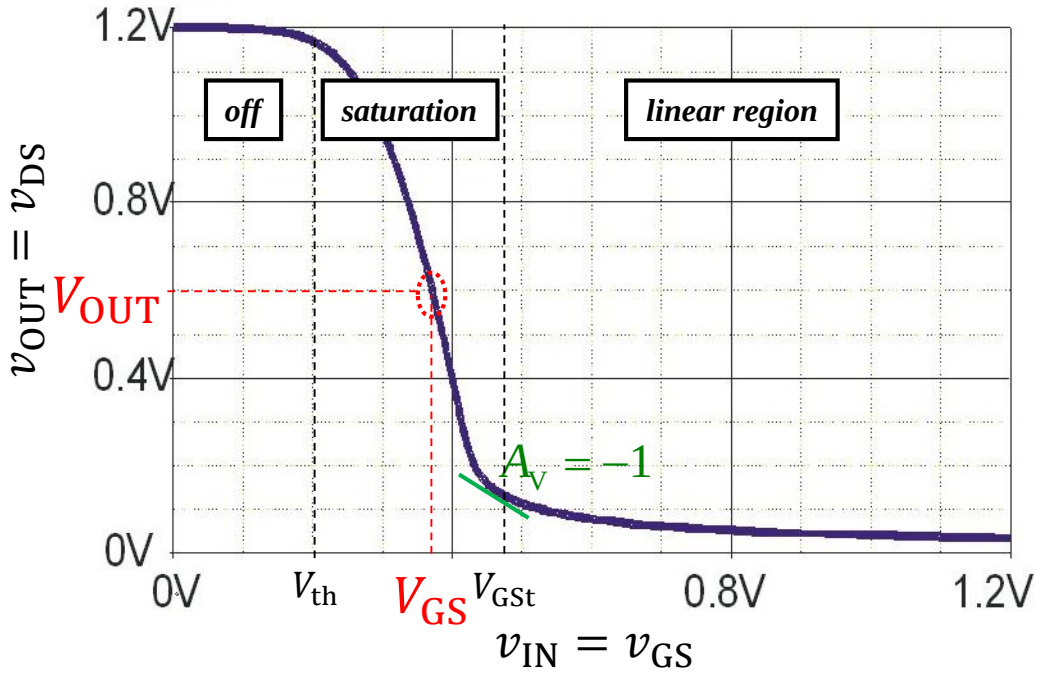
$$G_m = \frac{i_{out}}{v_{in}} = \frac{i_d}{v_{gs}}$$

$$A_V = \frac{v_{out}}{v_{in}} = -G_m R$$

Large-signal Behavior & Bias Point



inverter
with resistive load



$W/L = 10/10, \beta_0 = 250\ \mu A/V^2$
$V_{DD} = 1.2\ V, R = 100\ k\Omega$

operating / bias point
$V_{OUT}(V_{IN} = V_{GS})$
for analog operation
(in saturation)

high gain region:
$V_{th} < V_{GS} < V_{GSt}$

$$v_{IN} = v_{GS} \approx V_{IN} + v_{in}$$

$$v_{OUT} = v_{DS} = V_{DD} - R i_D$$

$$i_D = \frac{\beta}{2}(v_{IN} - V_{th})^2$$

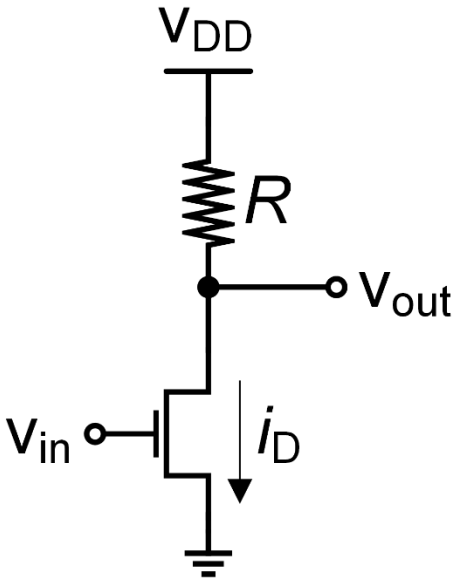
$$i_D = \beta\left(v_{GS} - V_{th} - \frac{1}{2}v_{DS}\right)v_{DS}$$

@ transition saturation to linear: $V_{OUT} = V_{DS} = V_{GSt} - V_{th}$

$$V_{OUT} = V_{GSt} - V_{th} = V_{DD} - R\frac{\beta}{2}(V_{GSt} - V_{th})^2$$

$$\rightarrow V_{GSt} = V_{th} - \frac{1}{\beta R} + \sqrt{\frac{1}{(\beta R)^2} + \frac{2V_{DD}}{\beta R}} \approx 0.47\ V$$

Small-Signal Behavior



inverter
with resistive load

$$A_V = \frac{v_{out}}{v_{in}}$$

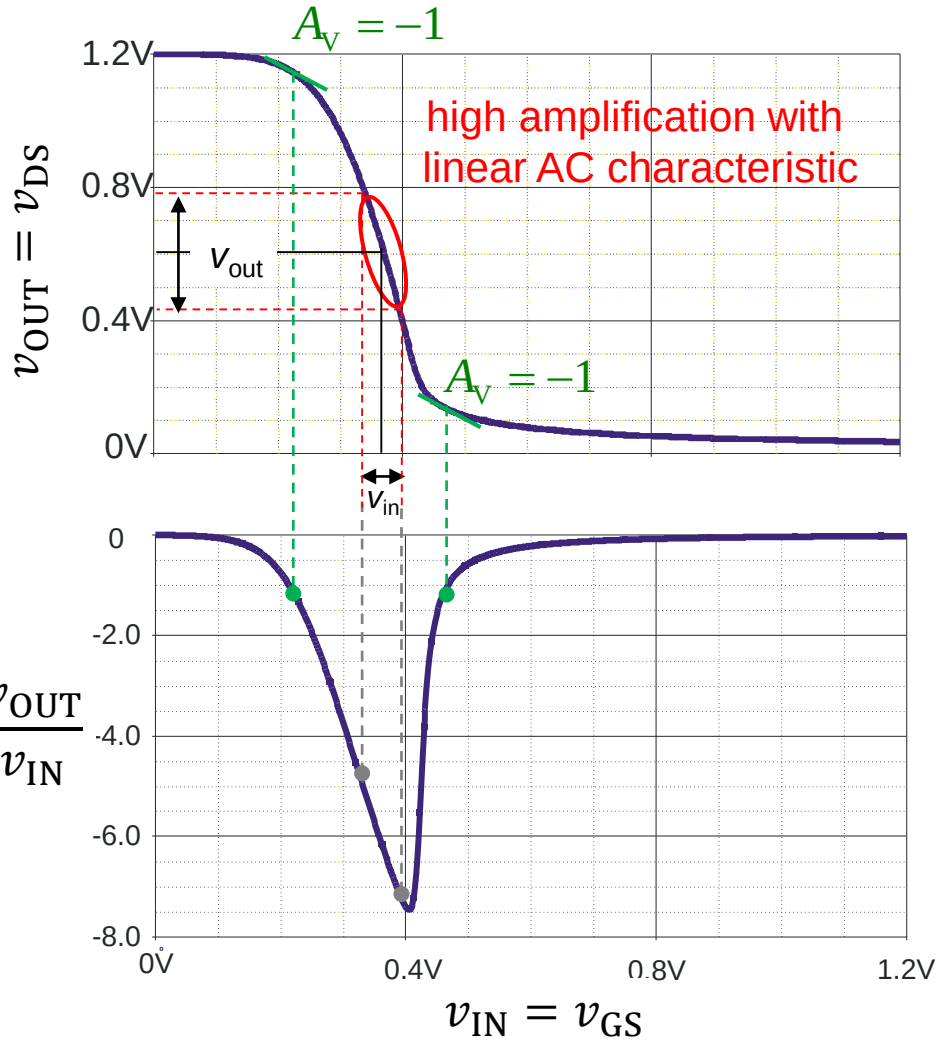
$$v_{OUT} = V_{DD} - i_D R$$

$$i_D = \frac{\beta}{2}(v_{GS} - V_{th})^2$$

$$= \frac{\beta}{2}(v_{IN} - V_{th})^2$$

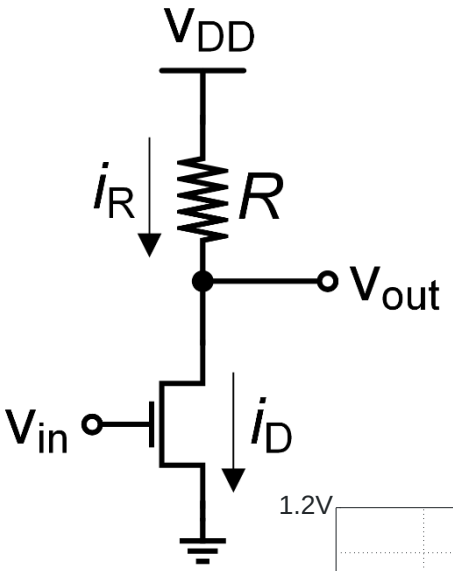
$$A_V = \left.\frac{\partial v_{OUT}}{\partial v_{IN}}\right|_{BP} = -\beta(V_{IN} - V_{th})R = -g_m R$$

$$A_{Vmax} = \left.\frac{\partial v_{OUT}}{\partial v_{IN}}\right|_{V_{IN}=V_{GSt}} = 1 - \sqrt{1 + 2\beta R V_{DD}} \approx -6.8$$



Resistive Load – Dynamic Range (Output Swing)

R=100k, $V_{DD} = 1.2V$, W/L=10/10



Output swing is a large-signal parameter!

$$V_{th} < v_{IN} < v_{DS} + V_{th} \Rightarrow v_{DS} > v_{IN} - V_{th}$$



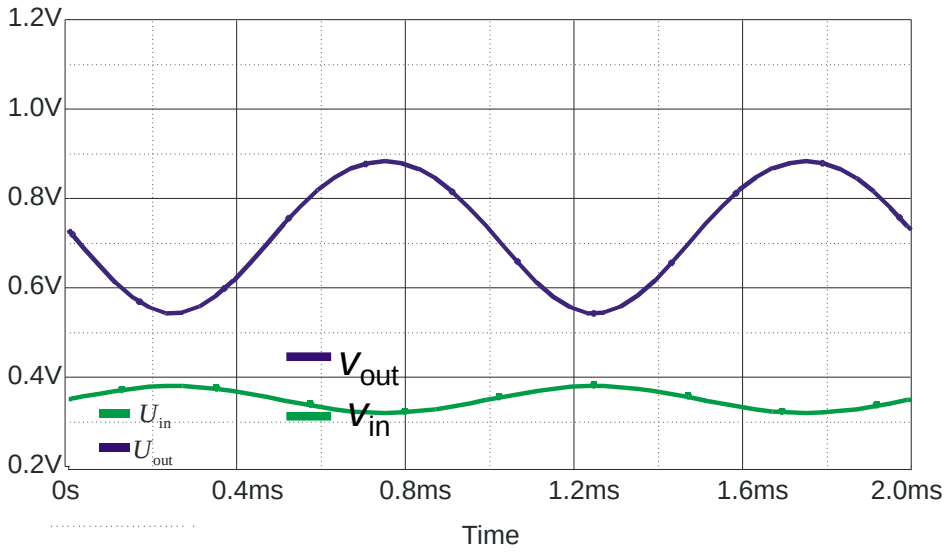
user: TMicroelectroncis

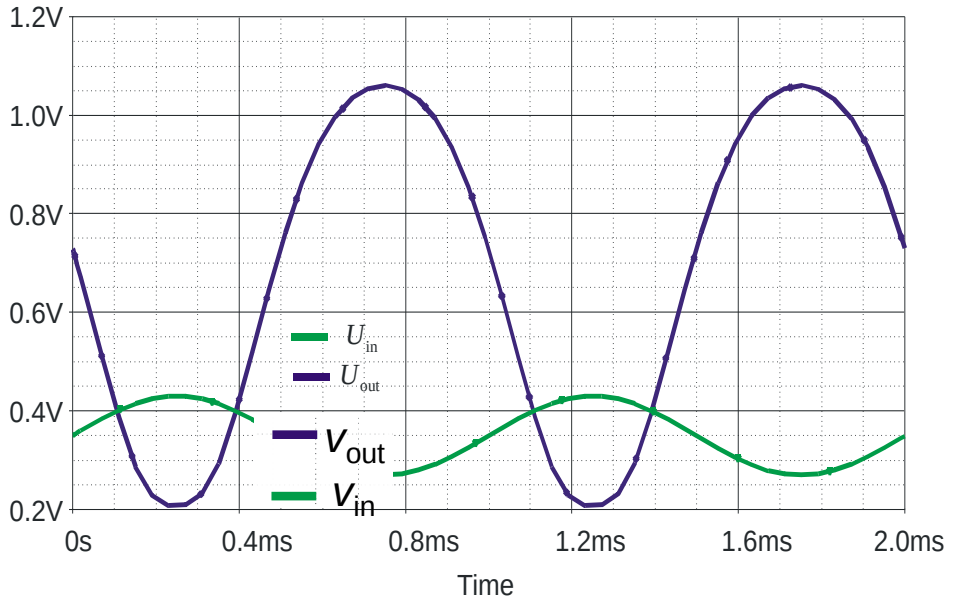
g_m depends on the operating point!

$$g_{m,sat} = \beta(V_{GS} - V_{th}) \Leftrightarrow g_{m,lin} = \beta V_{DS}$$

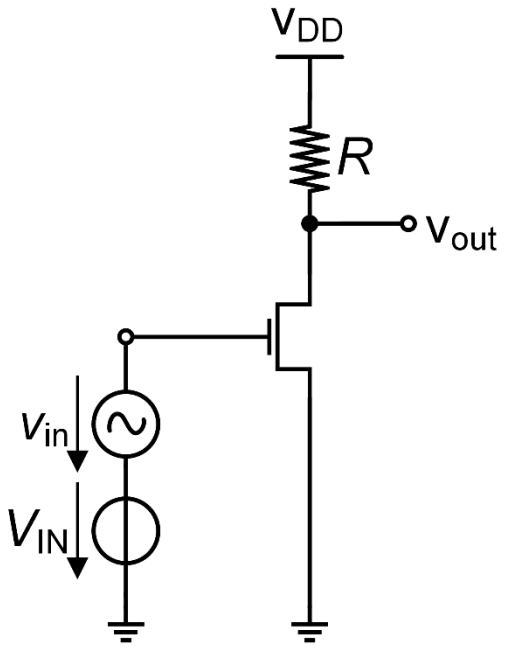
→ high input swing = strongly non-linear amplification

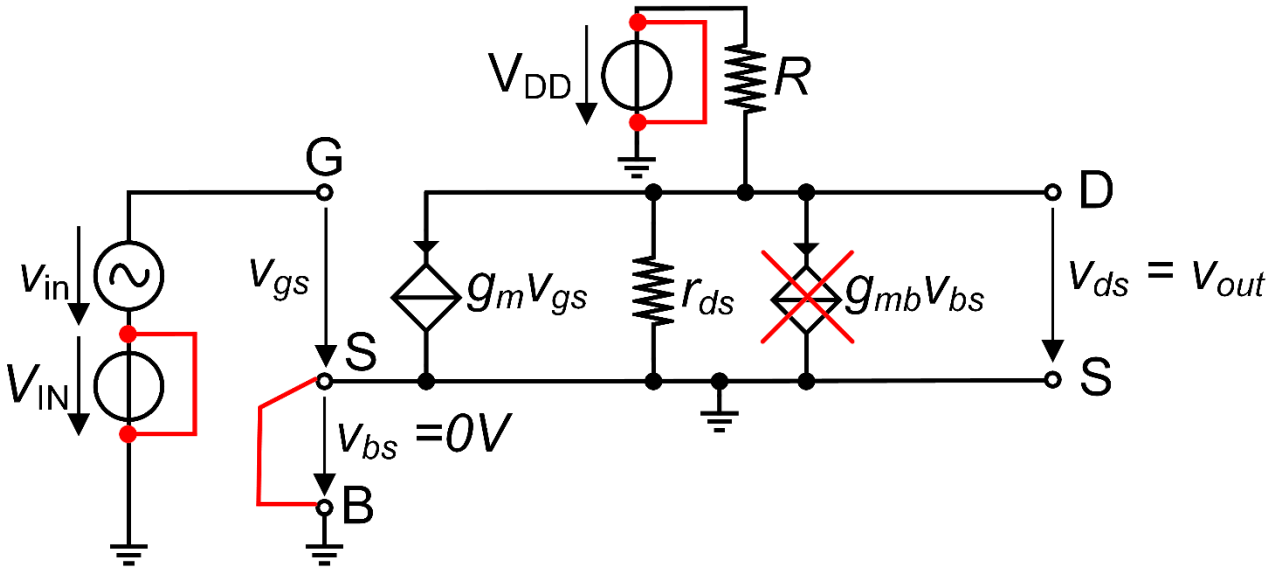
$$V_{IN} = 0.35\ V, v_{in} = 30\ mV/80\ mV$$





Amplification – Small-signal Equivalent Circuit



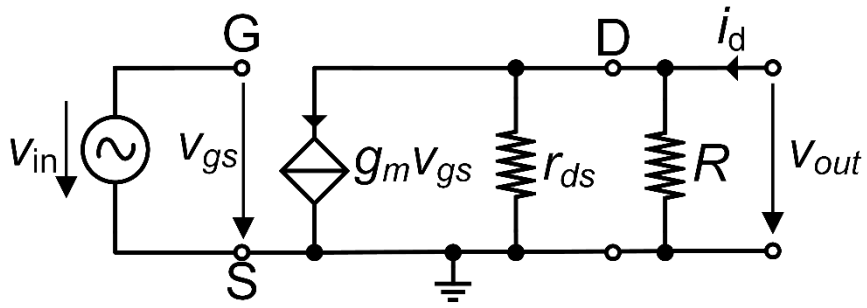


Replace all <u>constant</u> voltage-sources by short-circuits!

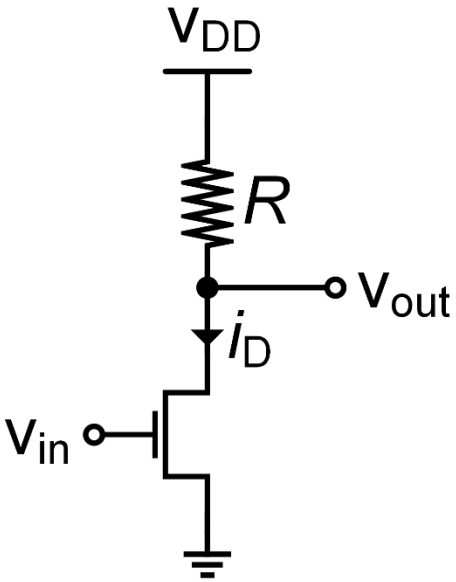
$(R_{\mathrm{IN}} = 0\ \Omega)$

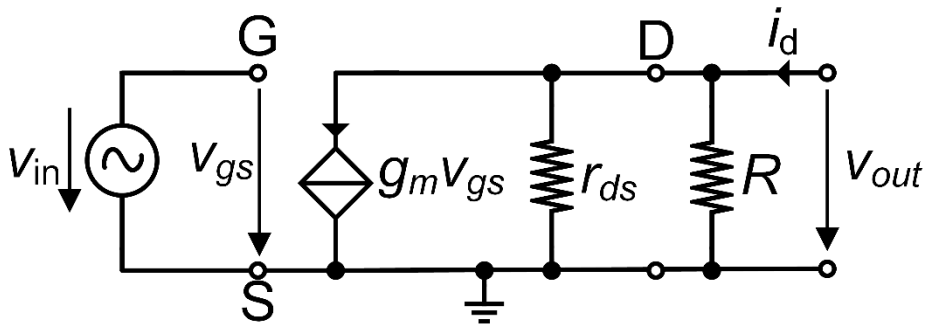
Replace all <u>constant</u> current-sources by open-circuits!

$(R_{\mathrm{IN}} = \infty\ \Omega)$



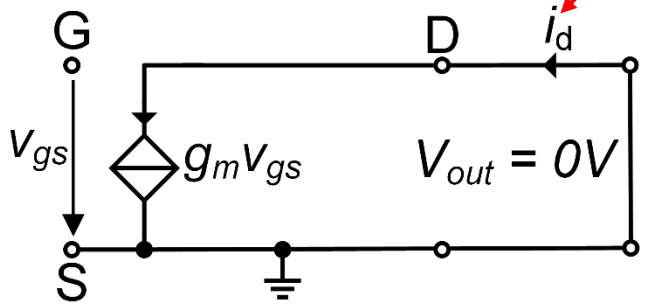
Approach I – Determine G_m & r_{out}



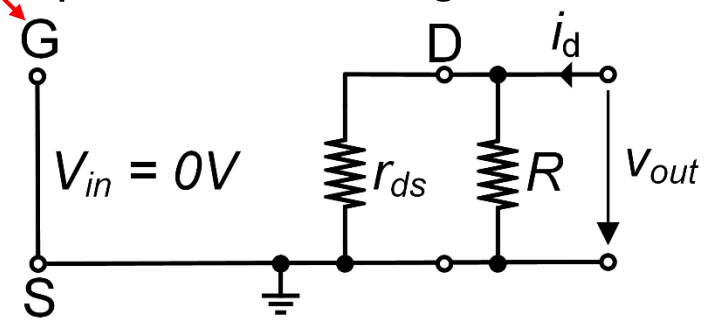


Why do we not "see" r_{ds} in the graphical discussion?

short-circuit current



open-circuit voltage



$$G_m = \left.\frac{i_d}{v_{in}}\right|_{v_{out}=0\,V}$$

$$r_{out} = \left.\frac{v_{out}}{i_d}\right|_{v_{in}=0\,V}$$

$$A = -g_m r_{out}$$

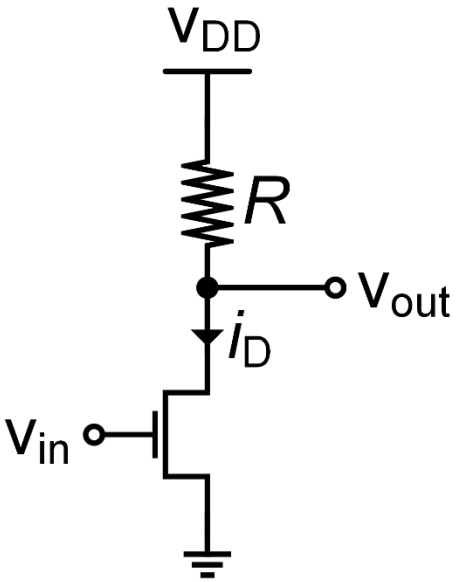
$$i_d = g_m v_{in}$$

$$G_m = g_m$$

$$v_{out} = i_d (R \,||\, r_{ds})$$

$$r_{out} = (R \,||\, r_{ds}) = \frac{R\, r_{ds}}{R + r_{ds}}$$

Resistive Load – Example



$$A = -g_m r_{\text{out}} \approx 6$$

$$\boldsymbol{R \uparrow \rightarrow V_{OUT} = ?}$$

process parameters

$$\beta_n = 370\ \frac{\mu\text{A}}{\text{V}^2}$$

$$\lambda = 0.02\ \frac{1}{\text{V}}$$

$$V_{\text{th}} = 0.19\ \text{V}$$

operating point

$$V_{\text{IN}} = 0.35\ \text{V}$$

$$V_{\text{OUT}} = 0.7\ \text{V}$$

$$V_{\text{SB}} = 0\ \text{V}$$

dimensioning

$$R = 100\ \text{k}\Omega$$

$$\frac{W}{L} = \frac{10\ \mu m}{10\ \mu\text{m}}$$

We thus get

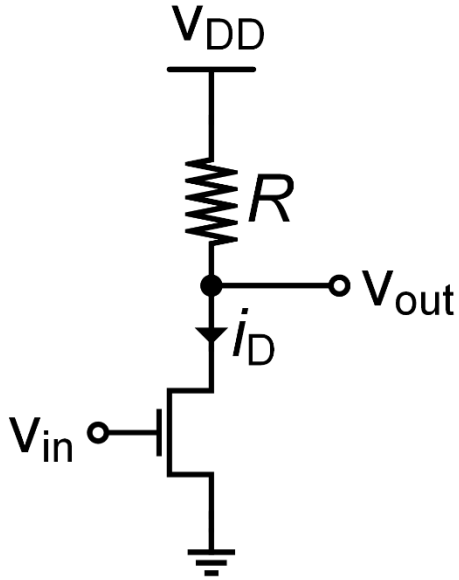
$$I_{\text{D}} = \frac{\beta}{2}(V_{\text{GS}} - V_{\text{th}})^2 \cdot (1 + \lambda V_{\text{DS}}) \Rightarrow I_{\text{D}} \approx 4.8\ \mu\text{A}$$

$$g_{\text{m}} = \beta(V_{\text{GS}} - V_{\text{th}}) \Rightarrow g_{\text{m1}} \approx 59\ \mu\text{S}$$

$$g_{\text{DS}} = \lambda I_{\text{D}} \Rightarrow g_{\text{DS1}} \approx 0.096\ \mu\text{S}$$

$$\Rightarrow r_{\text{DS1}} \approx 10\ \text{M}\Omega$$

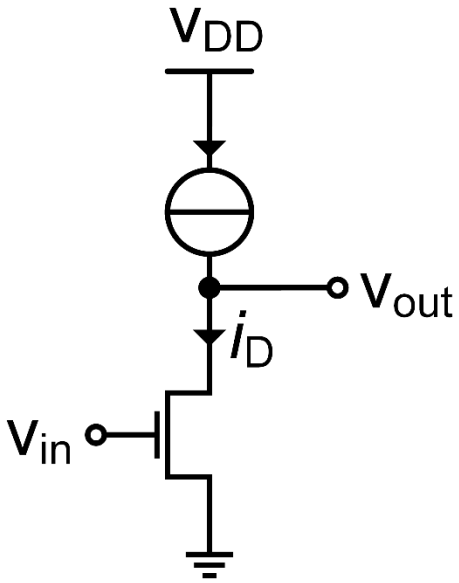
Biasing – Motivation



$$A_V = -g_\mathrm{m}(R \parallel r_\mathrm{ds})$$

Resistor *R* defines the gain of the inverting amplifier and needs to be large.

But it also defines the bias point of the output and can thus not be very large for reasonable currents.



$$A_V = -g_\mathrm{m} r_\mathrm{ds}$$

The load element needs to have a low *large-signal* resistance but a high *small-signal* resistance!

Amplifier should ideally be biased by a current source.

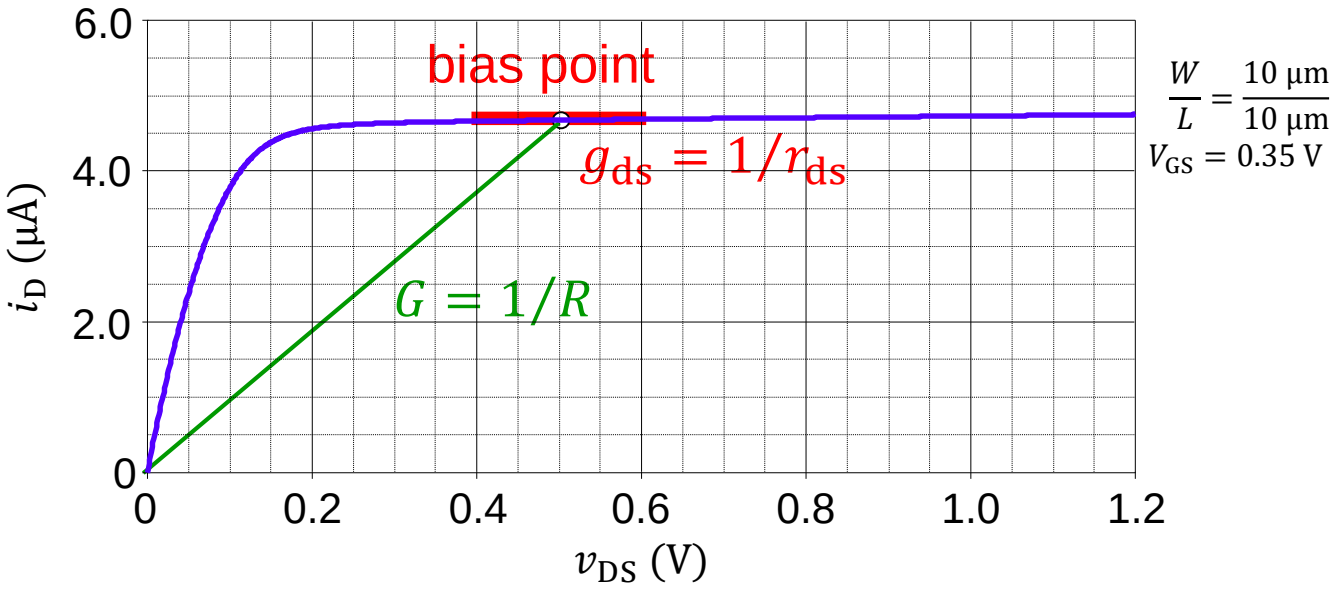
Motivation – Transistors as „Resistors“

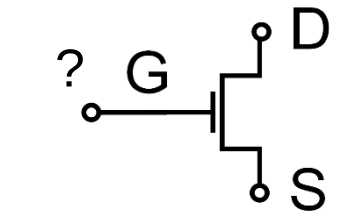
MOS transistors are voltage-controlled current sources.

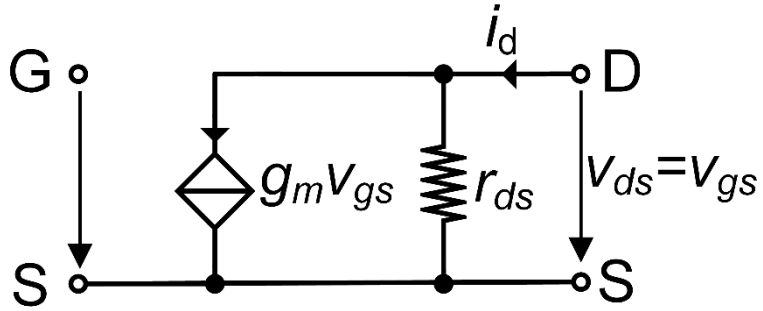
MOS transistors can behave as (small-signal) resistors.

The equivalent resistance depends on the operating point.

example: gate tied to fixed voltage







$$G = \frac{1}{R} = \frac{I_D}{V_{DS}}$$

$$\frac{\partial i_D}{\partial v_{DS}} = g_{ds} = \frac{1}{r_{ds}}$$

Fixed voltage?

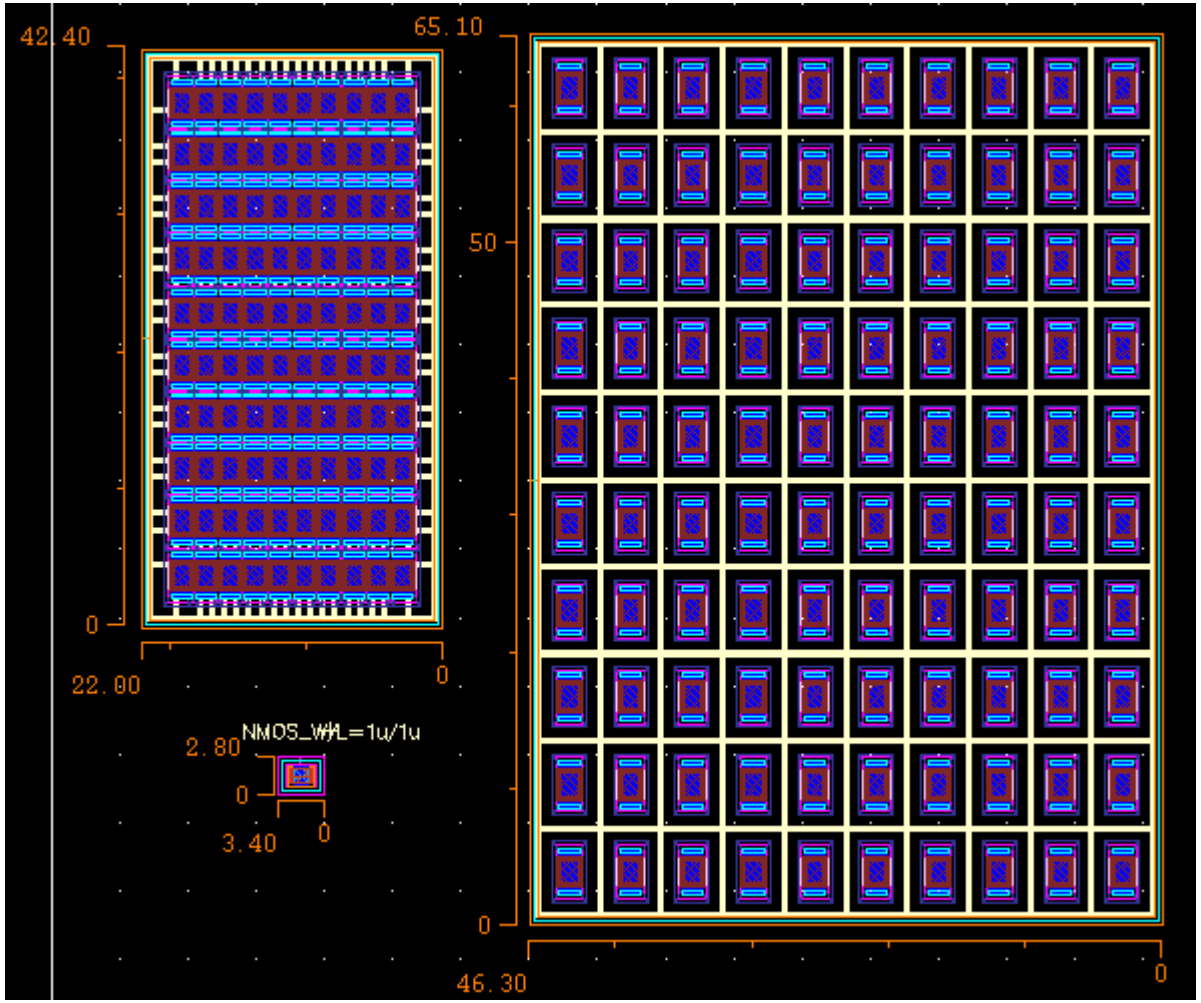
→ biasing / current mirror

Biasing – Motivation II

NMOS 1 µm / 1 µm

R = 100kΩ

The implementation of a reasonable resistance requires quite some area!



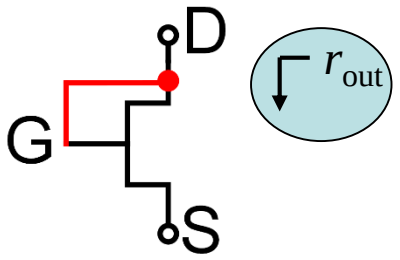
Kuhl, M. (2023). *Chapter 4: Single stage amplifiers* [Lecture slides]. Laboratory of Microelectronics, Department of Microsystems Engineering, IMTEK, University of Freiburg.

Overview

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  - Full Cascode CS Amplifier
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- Common-Gate (CG) Amplifier

Transistor as “NMOS Diode”

Bias the gate with the drain voltage! → two-terminal device



$$v_{DS} \geq v_{GS} - V_{th}$$

saturation $\quad i_D = \frac{\beta}{2}(v_{GS} - V_{th})^2$

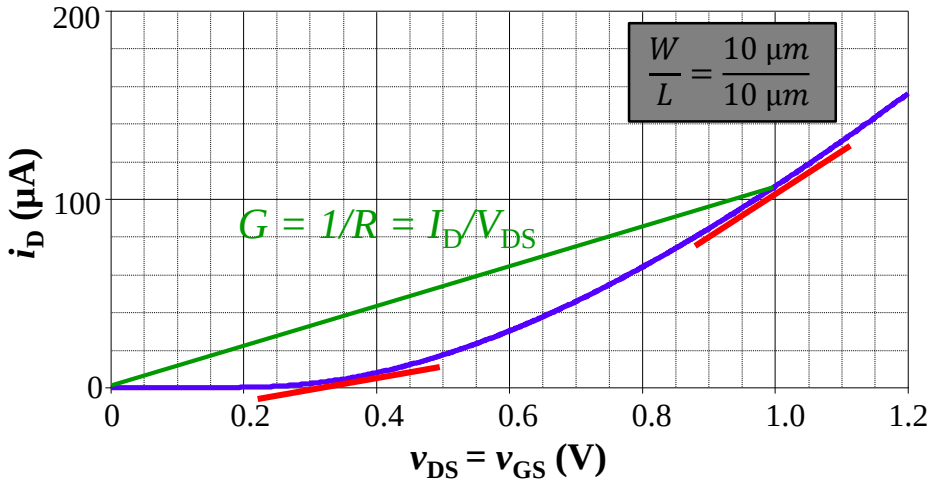
$$r_{out} = \frac{1}{g_m + g_{ds}}$$

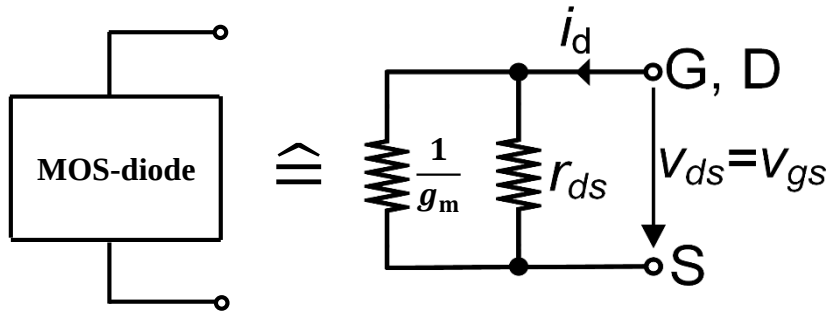
$V_{ds} = V_{gs}$

r_{out}

$$r_{out} = \frac{v_{out}}{i_{out}} = \frac{v_{ds}}{i_{out}}$$

$$i_{out} = g_m v_{gs} + g_{ds} v_{ds}$$
$$= g_m v_{ds} + g_{ds} v_{ds}$$





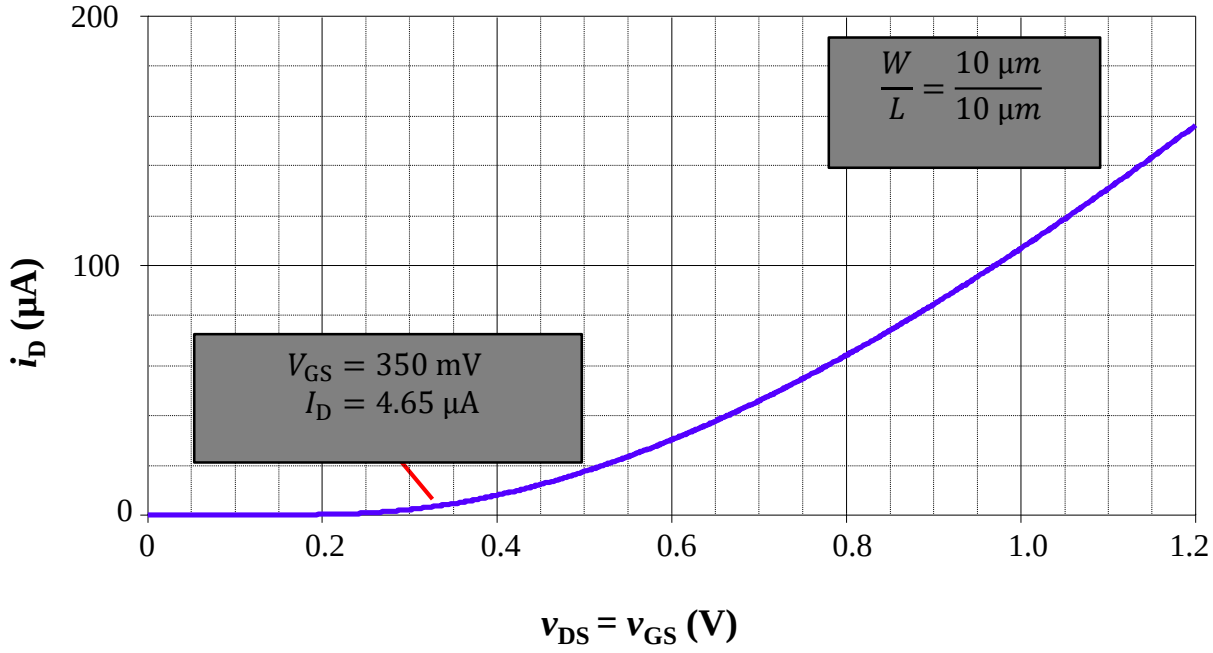
$$r_{out} = \frac{1}{g_m + g_{ds}}$$

Transistor as “nMOS Diode” – Example

$$g_\mathrm{m} = \sqrt{2\beta I_\mathrm{D}} = \sqrt{2\frac{W}{L}\beta_{0n} I_\mathrm{D}} = \sqrt{2\frac{10\ \mu m}{10\ \mu m}\,370\,\frac{\mu\mathrm{A}}{\mathrm{V}^2}\,4.65\ \mu A} \approx 58\ \mu\mathrm{S}$$

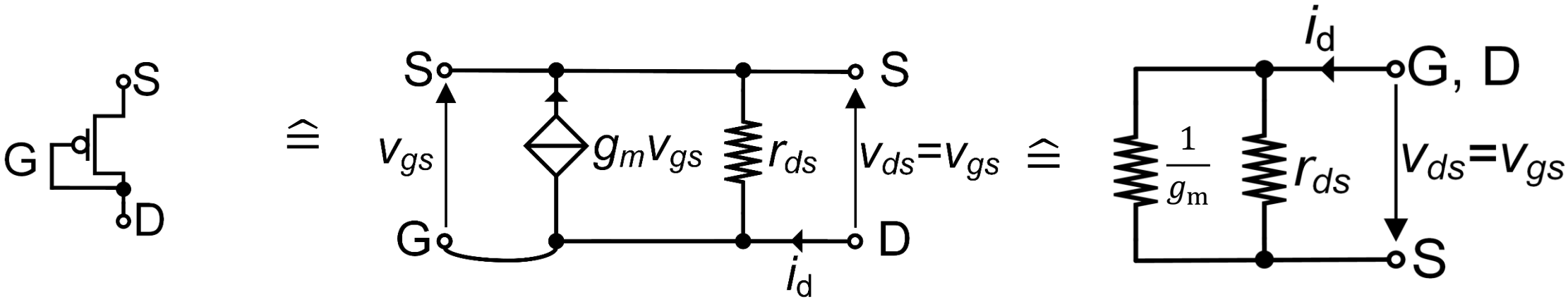
$$g_\mathrm{ds} \approx \lambda I_D = 0.027\ \mathrm{V}^{-1}\,4.65\ \mu\mathrm{A} \approx 0.123\ \mu\mathrm{S} \Rightarrow r_\mathrm{DS} \approx 8\mathrm{M}\Omega$$

rule of thumb
$g_{ds} < 0.1 \ldots 0.01 g_m$

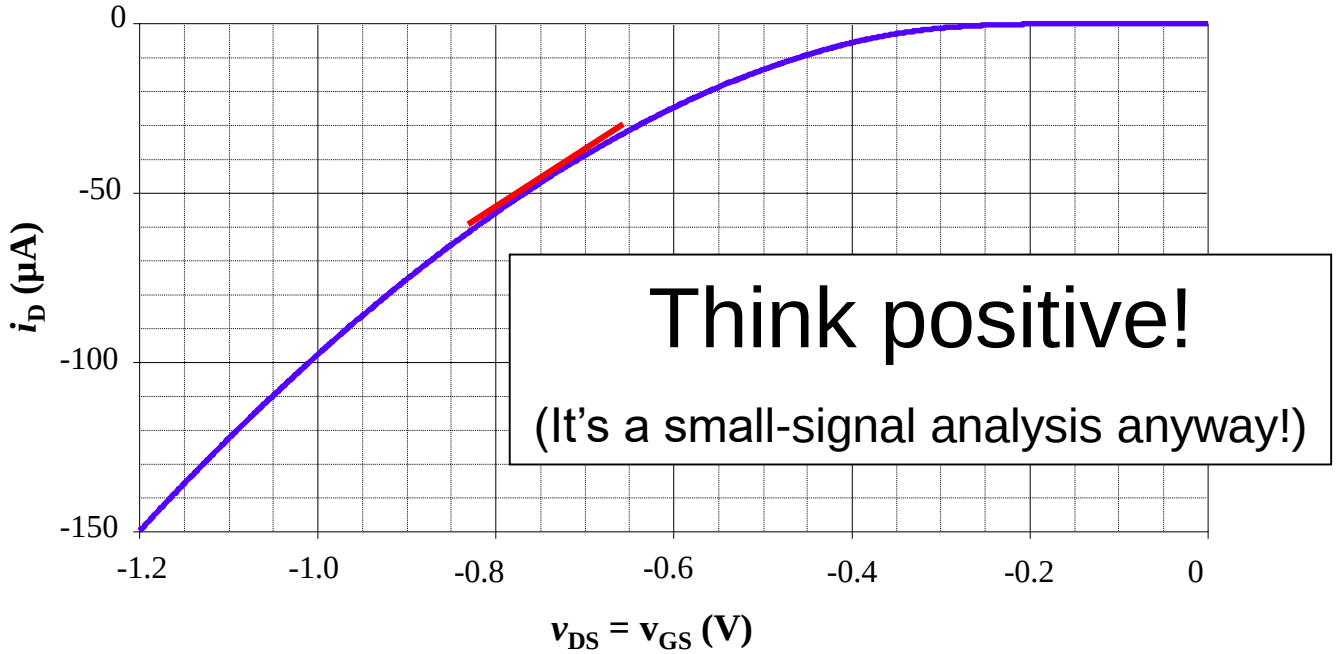


$$r_\mathrm{out} = \frac{1}{g_\mathrm{m} + g_\mathrm{ds}} \approx \frac{1}{g_\mathrm{m}} \approx 17\ \mathrm{k}\Omega$$

Transistor as “PMOS Diode”



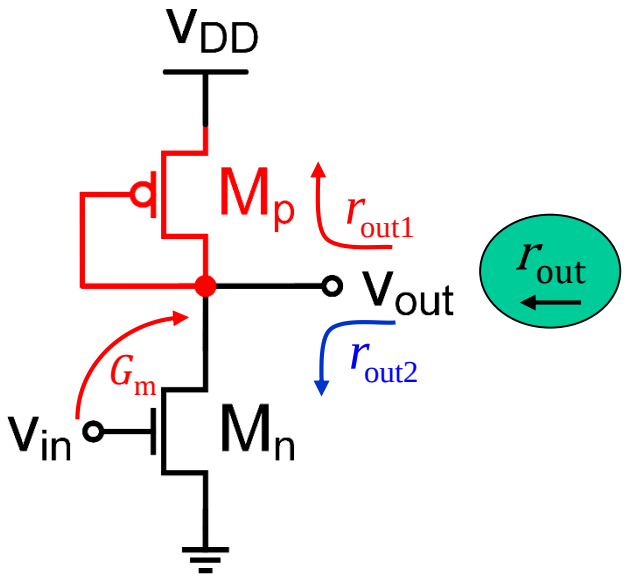
$$r_{out} = \frac{1}{g_m + g_{ds}} \approx \frac{1}{g_m}$$

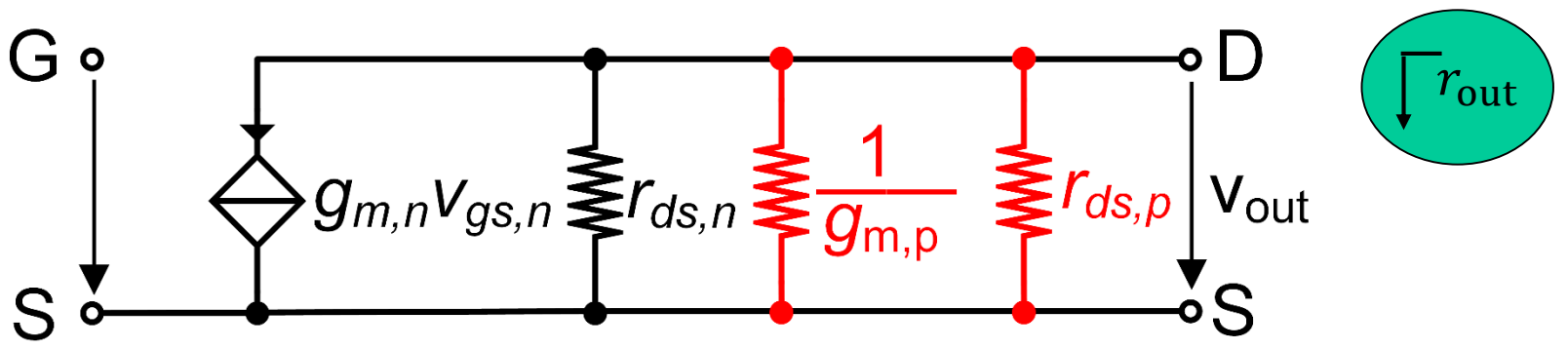


Think positive!

(It’s a small-signal analysis anyway!)

CS-Stage with Diode Load (PMOS) – AC Gain





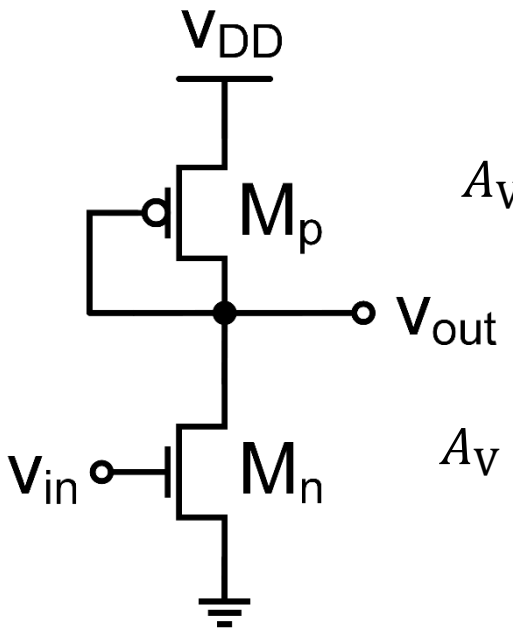
$$A_V = -G_m r_{out} = -\frac{g_{m,n}}{g_{m,p}}$$

$$G_m = g_{m,n} \qquad r_{out} = \frac{1}{g_{ds,n} + g_{m,p} + g_{ds,p}} \approx \frac{1}{g_{m,p}}$$

$$G_m = \left.\frac{i_{out}}{v_{in}}\right|_{v_{out} = 0\,V} \qquad r_{out} = \left.\frac{v_{out}}{i_{out}}\right|_{v_{in} = 0\,V}$$

$$g_{m,n} = \sqrt{2 I_D \beta_{0n} \left(\frac{W}{L}\right)_n} \qquad g_{m,p} = \sqrt{2 I_D \beta_{0p} \left(\frac{W}{L}\right)_p}$$

CS-Stage with Diode Load (PMOS) – AC Gain



$$A_V = -\frac{g_{m,n}}{g_{m,p}} \qquad g_{m,n} = \sqrt{2I_D\beta_{0n}\left(\frac{W}{L}\right)_n} \qquad g_{m,p} = \sqrt{2I_D\beta_{0p}\left(\frac{W}{L}\right)_p}$$

$$A_V = -\sqrt{\frac{\beta_{0n}\left(\frac{W}{L}\right)_n}{\beta_{0p}\left(\frac{W}{L}\right)_p}} = -\sqrt{\frac{\mu_n\left(\frac{W}{L}\right)_n}{\mu_p\left(\frac{W}{L}\right)_p}} \approx -\sqrt{\frac{4\left(\frac{W}{L}\right)_n}{\left(\frac{W}{L}\right)_p}}$$

The gain is independent of the bias current and voltage (as long as T_n is in saturation)!

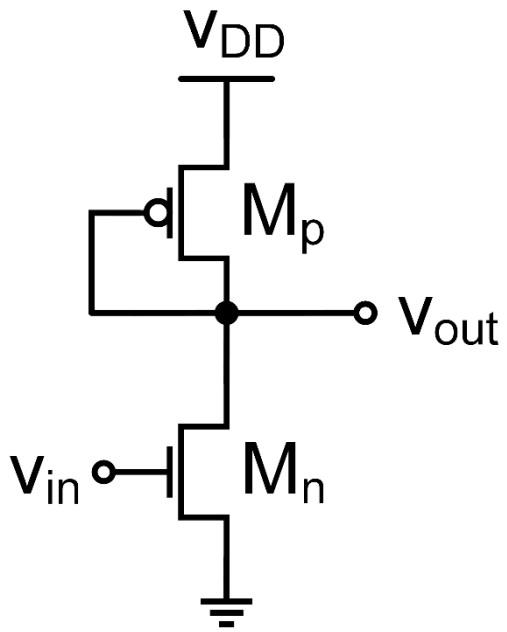
example:

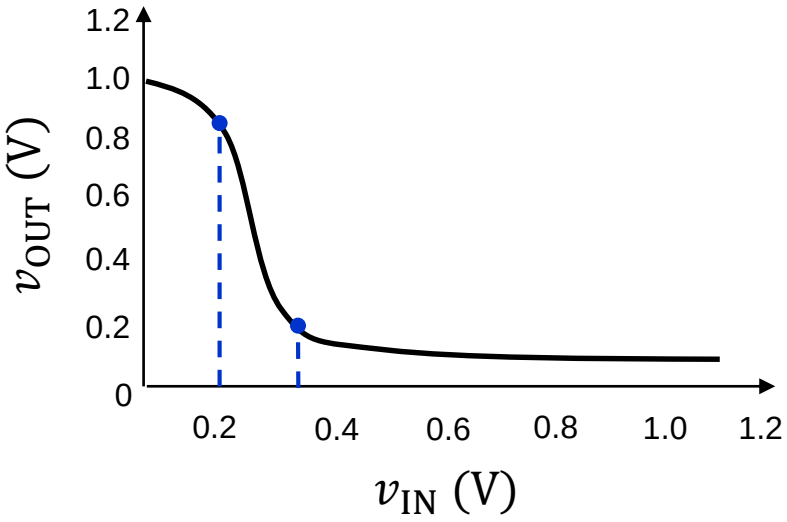
$$\left(\frac{W}{L}\right)_n = \left(\frac{10}{1}\right) \qquad \left(\frac{W}{L}\right)_p = \left(\frac{1}{1}\right)$$

$$|V_{th,p/n}| = 0.2\ V \qquad V_{DD} = 1.2\ V$$

$$A_V = -\sqrt{\frac{4\left(\frac{W}{L}\right)_n}{\left(\frac{W}{L}\right)_p}} = -\sqrt{\frac{4\left(\frac{10}{1}\right)_n}{\left(\frac{1}{1}\right)_p}} \approx -6.5$$

CS-Stage with Diode Load (PMOS)–Transfer Characteristic & Signal Swing





$$v_{\mathrm{OUT,max}} = V_{\mathrm{DD}} - V_{\mathrm{th,p}} \qquad v_{\mathrm{OUT,min}} = v_{\mathrm{DSn,sat}} = v_{\mathrm{GSn,eff}}$$

$$\boxed{133\ \mathrm{mV} \leq v_{\mathrm{OUT}} \leq 1\ \mathrm{V}}$$

@ transition saturation to linear: $V_{\mathrm{OUT}} = V_{\mathrm{DSn}} = V_{\mathrm{GSt}} - V_{\mathrm{th,n}}$

$$V_{\mathrm{OUT}} = V_{\mathrm{GSt}} - V_{\mathrm{th,n}} = V_{\mathrm{DD}} - |V_{\mathrm{GS,p}}| = V_{\mathrm{DD}} - \sqrt{\frac{2I_{\mathrm{Dp}}}{\beta_{\mathrm{p}}}} - V_{\mathrm{th,p}}$$

$$\text{with } I_{\mathrm{Dp}} = I_{\mathrm{Dn}}$$

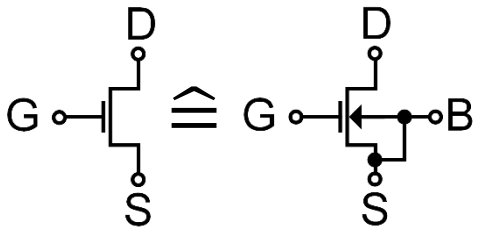
$$\rightarrow V_{\mathrm{GSt}} = \frac{V_{\mathrm{DD}} - V_{\mathrm{th,p}}}{1 + \sqrt{\beta_{\mathrm{n}}/\beta_{\mathrm{p}}}} + V_{\mathrm{th,n}}$$

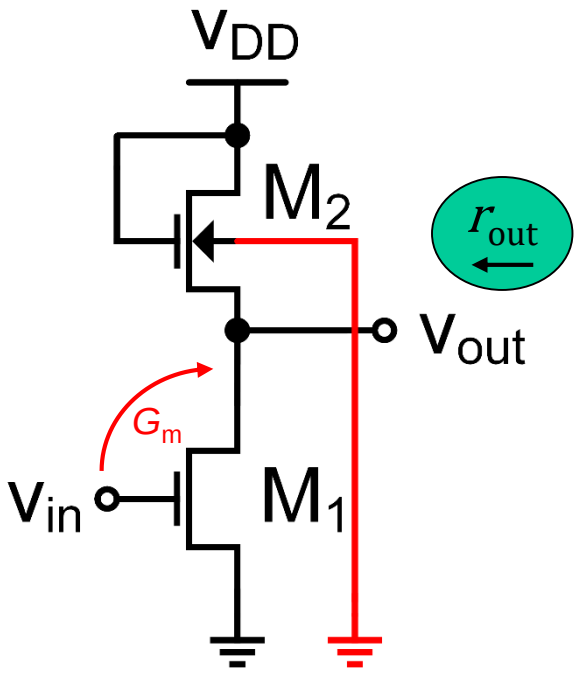
$$= \frac{V_{\mathrm{DD}} - V_{\mathrm{th,p}}}{1 + A_{\mathrm{V}}} + V_{\mathrm{th,n}}$$

$$\approx 333\ \mathrm{mV}$$

$$\boxed{200\ \mathrm{mV} \leq V_{\mathrm{GSn}} \leq 333\ \mathrm{mV}}$$

CS-Stage with Diode Load (NMOS) – AC Gain





$$A_V = -G_m r_{out}$$

$$G_m = g_{m1}$$

$$r_{out} = \frac{1}{g_{ds1} + g_{ds2} + g_{m2} + g_{mb2}} \approx \frac{1}{g_{m2} + g_{mb2}}$$

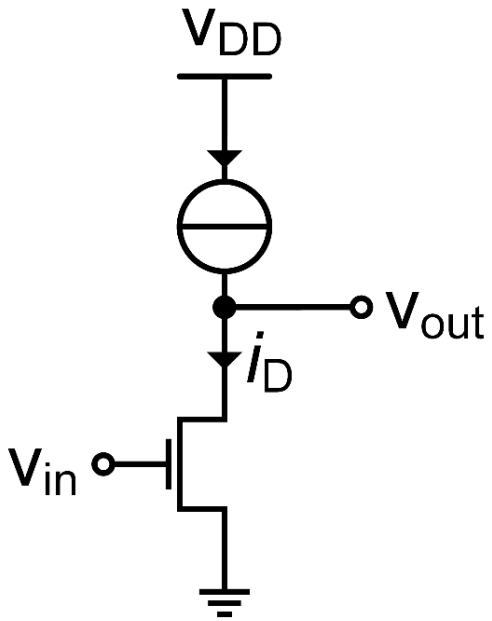
$$A_V \approx -\frac{g_{m1}}{g_{m2} + g_{mb2}} = -\frac{g_{m1}}{g_{m2}} \frac{1}{1+\eta}$$

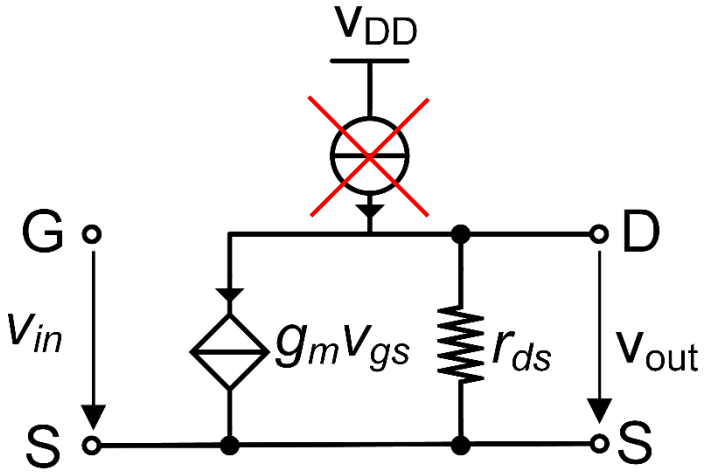
→ Body-effect decreases the gain!

Overview

- **Common-Source (CS) Amplifier**
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CS-Stage with Ideal Current Source Load





task:
Implement a real current source as load!

result: same gain as CS stage with resistive load $R \gg r_{\mathrm{ds}}$

$$A_{\mathrm{V}} = -G_{\mathrm{m}} r_{\mathrm{out}}$$

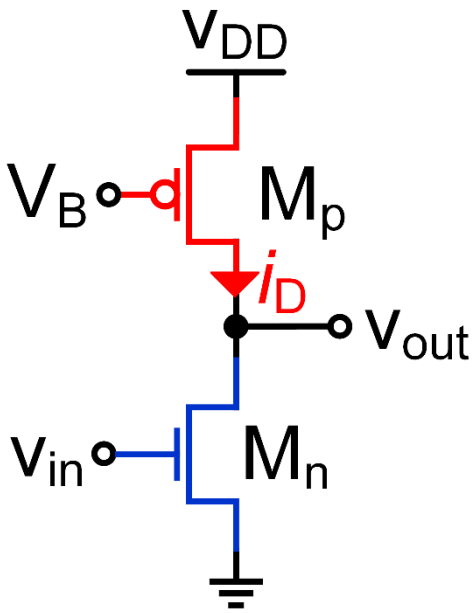
$$G_{\mathrm{m}} = g_{\mathrm{m}}$$

$$r_{\mathrm{out}} = r_{\mathrm{out1}} \parallel r_{\mathrm{out2}} = \infty \parallel r_{\mathrm{ds}} = r_{\mathrm{ds}}$$

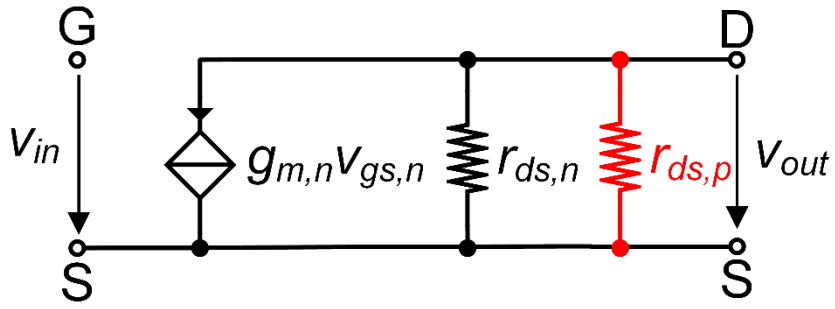
$$A_{\mathrm{V}} = -g_{\mathrm{m}} r_{\mathrm{ds}} = -\sqrt{2\beta \left(\frac{W}{L}\right) I_D} \frac{1}{\lambda I_D}$$

intrinsic transistor gain

CS-Stage with Current Source Load (PMOS)

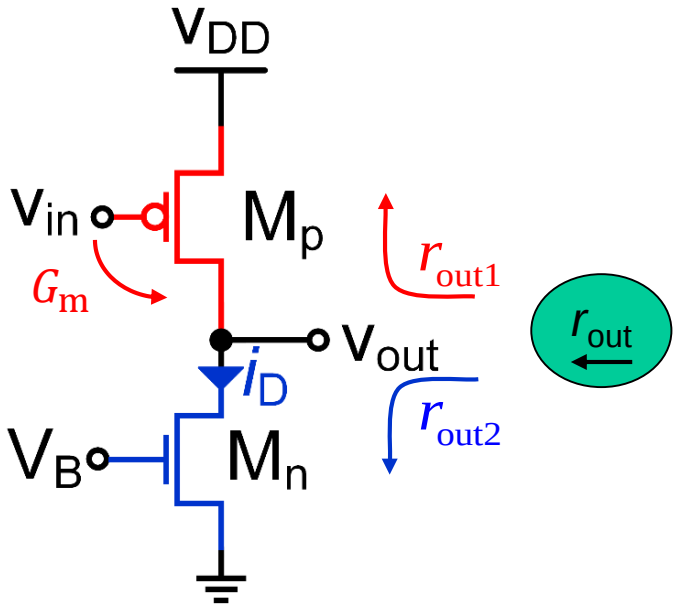


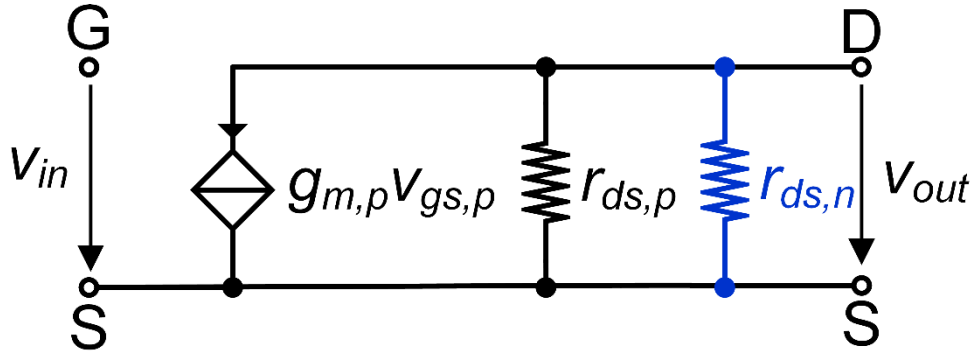
nMOS is driven
→ pMOS as load



$$A_v = -G_m r_{out}$$
$$= -g_{m,n}(r_{ds,n} || r_{ds,p}) = -\frac{g_{m,n}}{g_{ds,n} + g_{ds,p}}$$

CS-Stage with Current Source Load (NMOS)





pMOS is driven
→ nMOS as load

$$A_v = -G_m r_{out}$$
$$= -g_{m,p}(r_{ds,n} || r_{ds,p}) = -\frac{g_{m,p}}{g_{ds,n} + g_{ds,p}}$$

CS-Stage with Current Source Load – Signal Swing

input dynamic range

$v_{DS,n} = v_{OUT}$ $\quad$ $v_{GS,n} = v_{IN}$

saturation limit: $v_{DS,n} \geq v_{GS,n} - V_{th,n}$

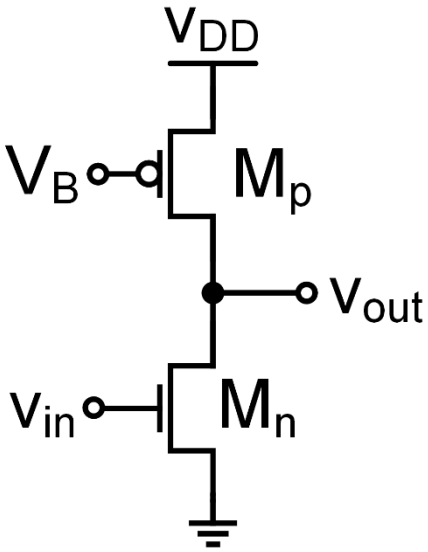
$$v_{OUT,min} \geq v_{IN(max)} - V_{th,n}$$

$$v_{IN(max)} \leq v_{OUT,min} + V_{th,n}$$

$$\leq v_{GSeff,n} + V_{th,n}$$

$$\Rightarrow v_{IN(min)} = v_{GSeff,n} + V_{th,n}$$

$$\boldsymbol{v_{GSeff,n} + V_{th,n} < v_{IN} < v_{GSeff,n} + V_{th,n}}$$



output dynamic range

$v_{SD,p} = V_{DD} - v_{OUT}$ $\quad$ $v_{SG,p} = V_{DD} - V_B$

saturation limit: $v_{SD,p} \geq v_{SG,p} - |V_{th,p}|$

$$V_{DD} - v_{OUT} \geq V_{DD} - V_B - |V_{th,p}|$$

$$\Rightarrow v_{OUT,max} \leq V_B + |V_{th,p}| = V_{DD} - v_{SGeff,p}$$

$$\Rightarrow v_{OUT,min} \geq \begin{cases} v_{IN,min} \\ v_{IN,max} \end{cases} - V_{th,n} = v_{GSeff,n}$$

$$\boldsymbol{v_{GSeff,n} < v_{OUT} < V_{DD} - |v_{GSeff,p}|}$$

requirement:

All transistors operate in saturation!

$$v_{GS,n} > V_{th,n}$$

$$v_{DS,n} > v_{GS,n} - V_{th,n}$$

$$|v_{GS,p}| > |V_{th,p}|$$

$$|v_{DS,p}| > |v_{GS,p}| - |V_{th,p}|$$

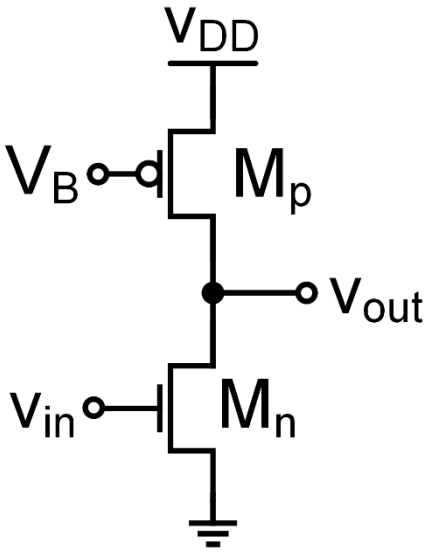
CS-Stage with Current Source Load – Signal Swing

input dynamic range

$$v_{\mathrm{GSeff,n}} + V_{\mathrm{th,n}} < v_{\mathrm{IN}} < v_{\mathrm{GSeff,n}} + V_{\mathrm{th,n}}$$

$$0.15\ \mathrm{V} + 0.2\ \mathrm{V} < v_{\mathrm{IN}} < 0.15\ \mathrm{V} + 0.2\ \mathrm{V}$$

$$0.35\ \mathrm{V} < v_{\mathrm{IN}} < 0.35\ \mathrm{V}$$



output dynamic range

$$v_{\mathrm{GSeff,n}} < v_{\mathrm{OUT}} < V_{\mathrm{DD}} - v_{\mathrm{SGeff,p}}$$

$$0.15\ \mathrm{V} < v_{\mathrm{OUT}} < 1.2\ \mathrm{V} - 0.15\ \mathrm{V}$$

$$0.15\ \mathrm{V} < v_{\mathrm{OUT}} < 1.05\ \mathrm{V}$$

example:

$$V_{\mathrm{DD}} = 1.2\ \mathrm{V}$$

$$V_{\mathrm{th,n}} = |V_{\mathrm{th,p}}| = 0.2\ \mathrm{V}$$

$$v_{\mathrm{GSeff,n}} = |v_{\mathrm{GSeff,p}}| = 0.15\ \mathrm{V}$$

First Summary

- CS with diode-connected transistor load
  - low small-signal gain
  - strong input & week load device are required for high gain
- CS amplifier with ideal current source load
  - intrinsic transistor gain (highest possible gain of a single transistor)
  - gain is limited by the output resistance of the transistor (= input device)
- CS amplifier with real current source load
  - high gain
  - limited input & output dynamic range (signal swing)

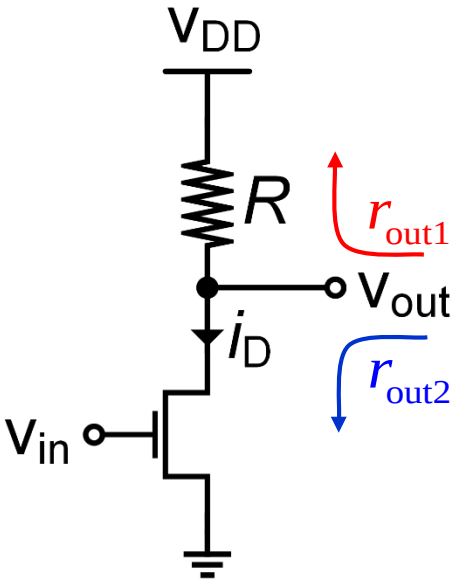
Literature

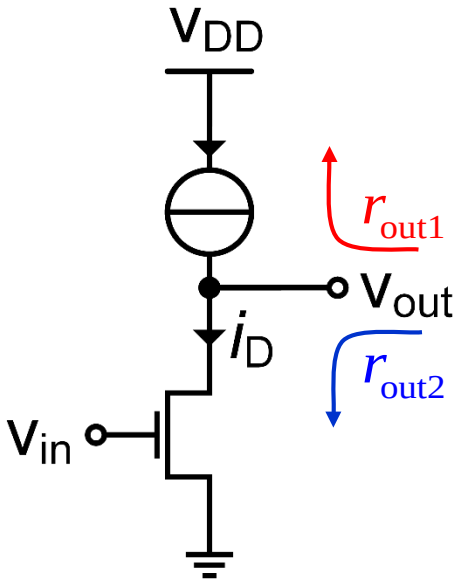
- Razavi, chapters 3.2.2, page 53-58
- Razavi, chapters 3.2.3, page 58-59

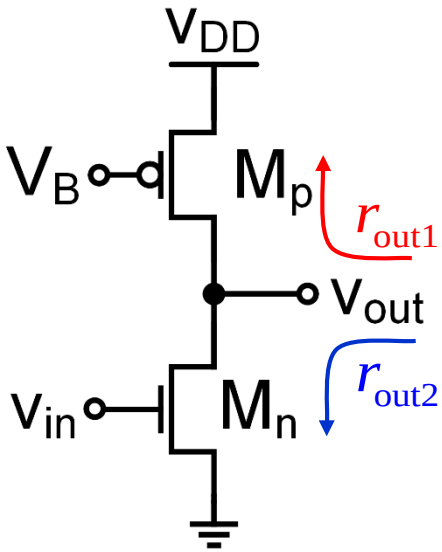
Overview

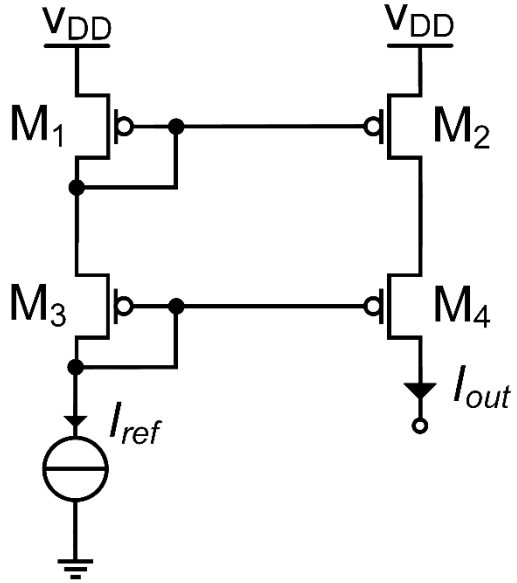
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Better Load Element?









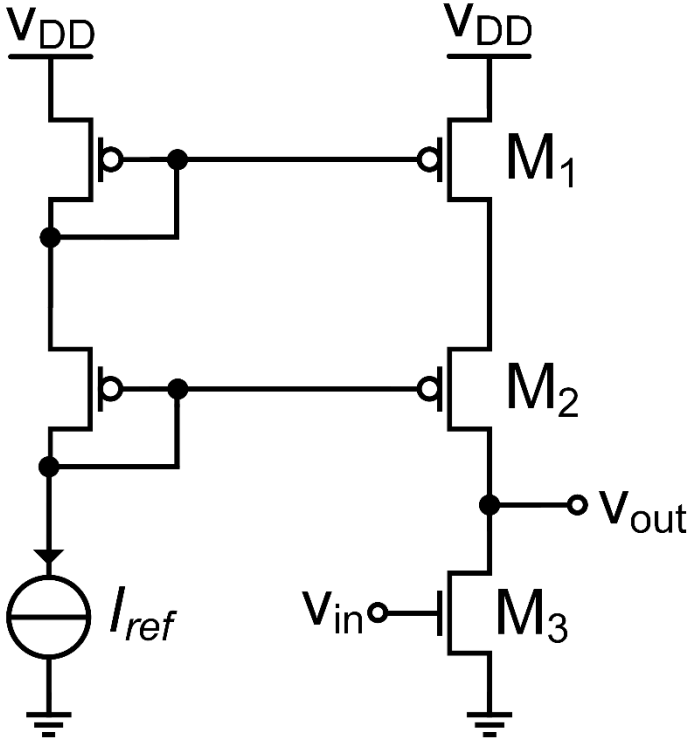
MOS cascode current source

Recollect:

$$A_{\mathrm{v}} = -g_{\mathrm{m}}(r_{\mathrm{out1}} \parallel r_{\mathrm{out2}})$$

How can we boost r_{out} further?

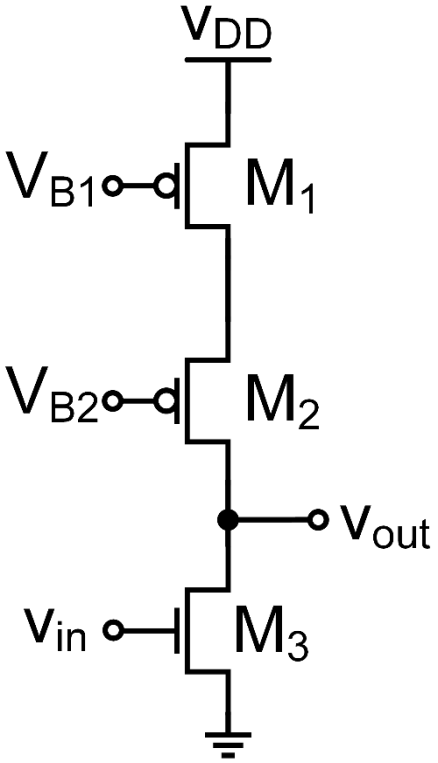
CS Amplifier with Cascode Current Source Load



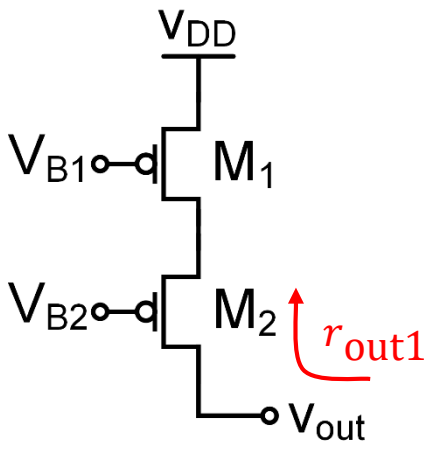
small-signal analysis

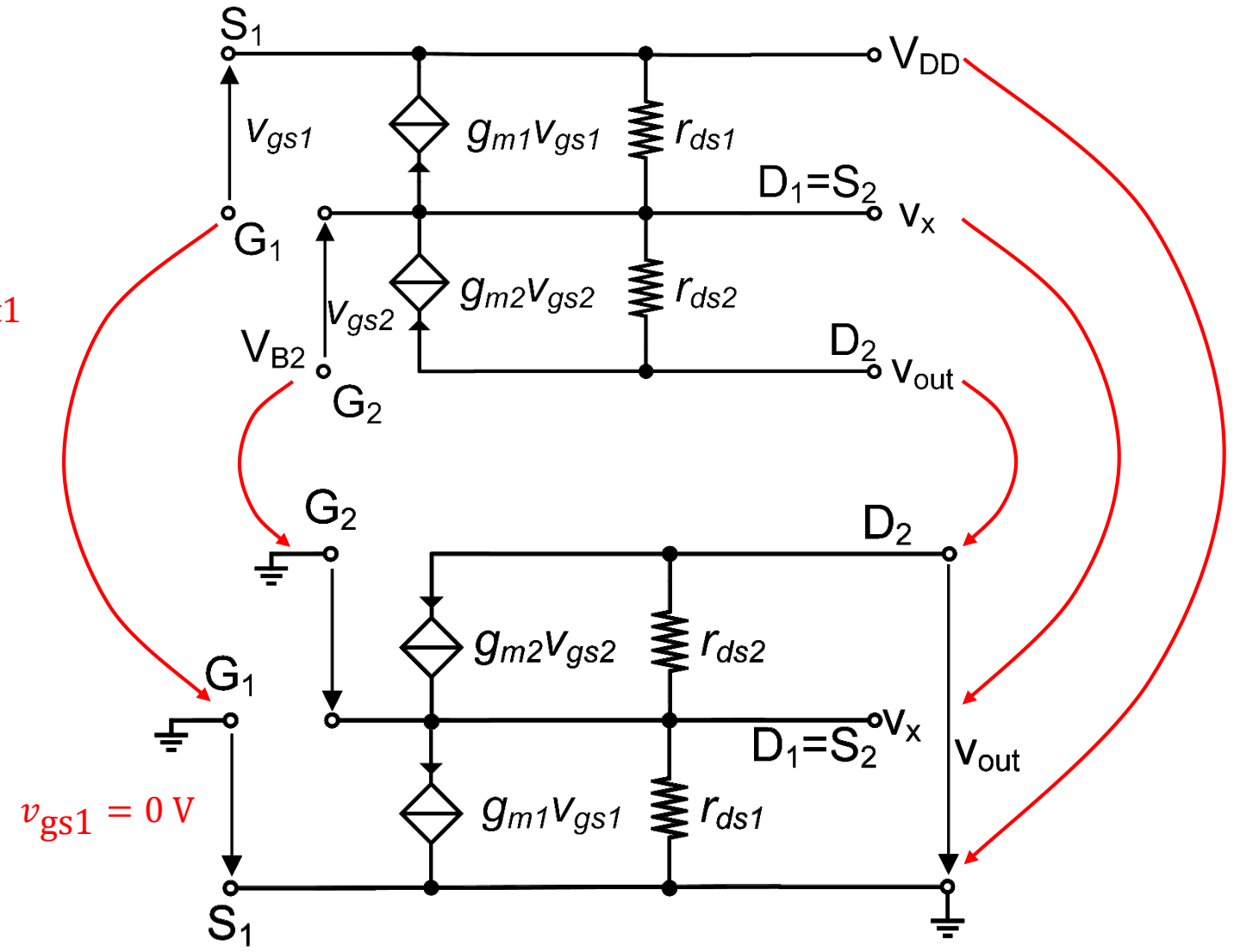
⇒

The reference branch can be removed for the small-signal analysis!
It's task is to generate (constant) bias voltages that are shorted in the small-signal analysis anyway!



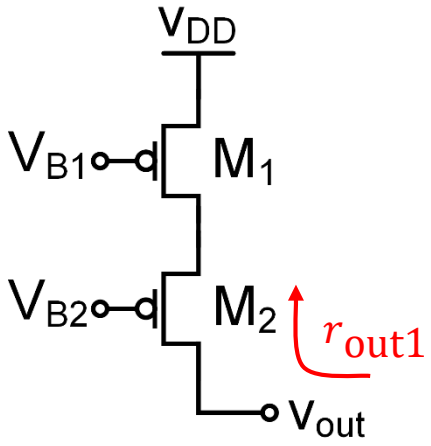
Cascode Current Source – Recap

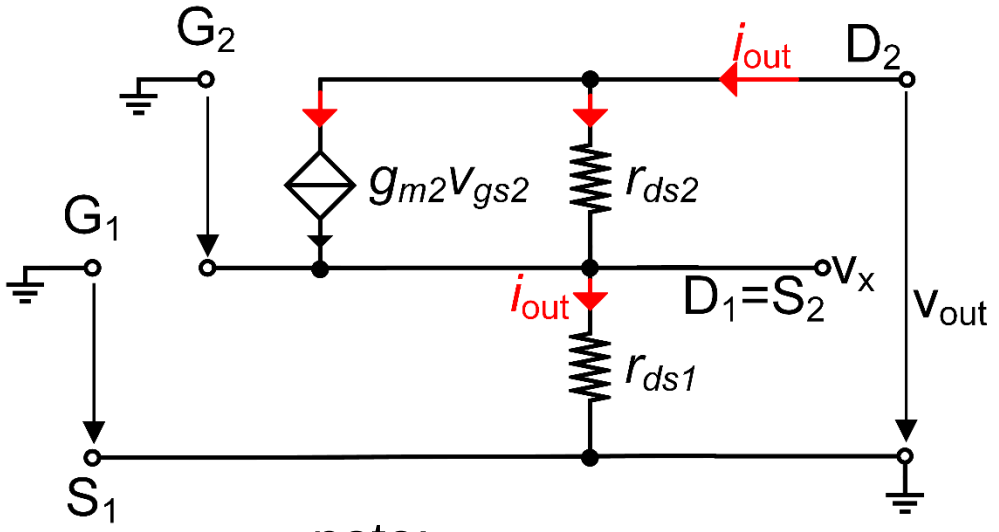




We like to see ground or virtual ground at the lower part of the schematic!

Cascode Current Source – Output Resistance





note: $v_{gs2} = -v_x$

$$r_{out} = \left.\frac{v_{out}}{i_{out}}\right|_{v_{in}=0\,V}$$

$$v_{out} = v_x + i_{r2} r_{ds2}$$

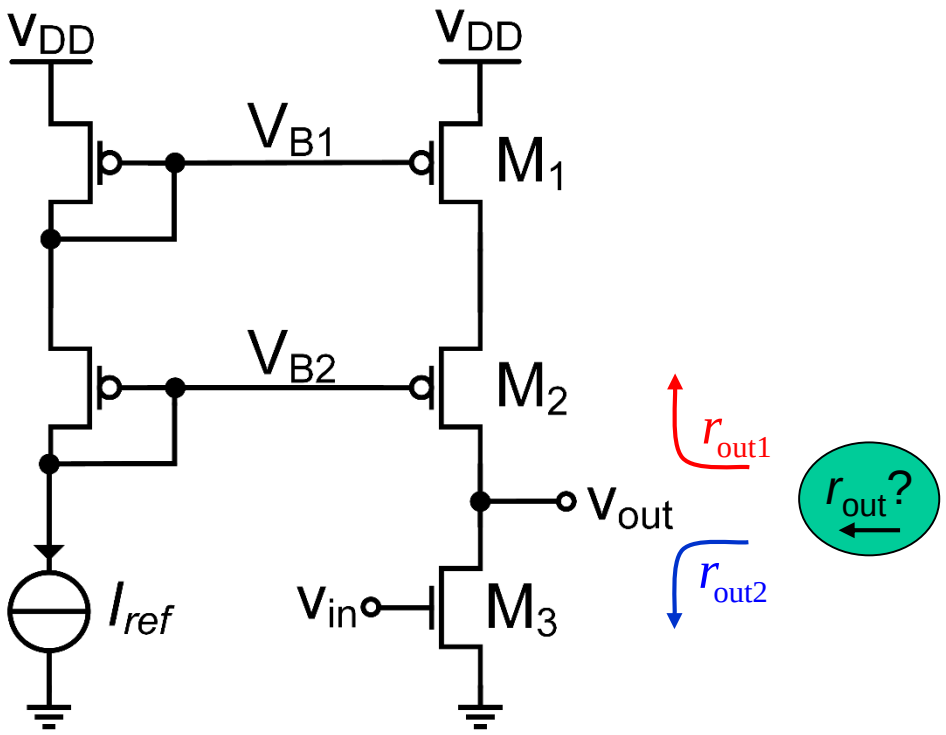
$$v_x = r_{ds1} i_{out}$$

$$i_{r2} = i_{out} - g_{m2} v_{gs2}$$
$$= i_{out} + g_{m2} v_x = i_{out} + g_{m2} r_{ds1} i_{out}$$

$$v_{out} = i_{out} r_{ds1} + (i_{out} + g_{m2} r_{ds1} i_{out}) r_{ds2}$$
$$= i_{out}(r_{ds1} + r_{ds2} + g_{m2} r_{ds2} r_{ds1})$$

$$\boxed{\frac{v_{out}}{i_{out}} = r_{out} \approx g_{m2} r_{ds2} r_{ds1}}$$

CS Amplifier with Cascode Current Source Load – Small-Signal Gain



$$A_V = -g_{m3} r_{out}$$

$$r_{out} = r_{out1} || r_{out2}$$

$$r_{out1} \approx g_{m2} r_{ds2} r_{ds1}$$

$$r_{out2} = r_{ds3}$$

$$r_{out} \approx r_{out2} = r_{ds3}$$

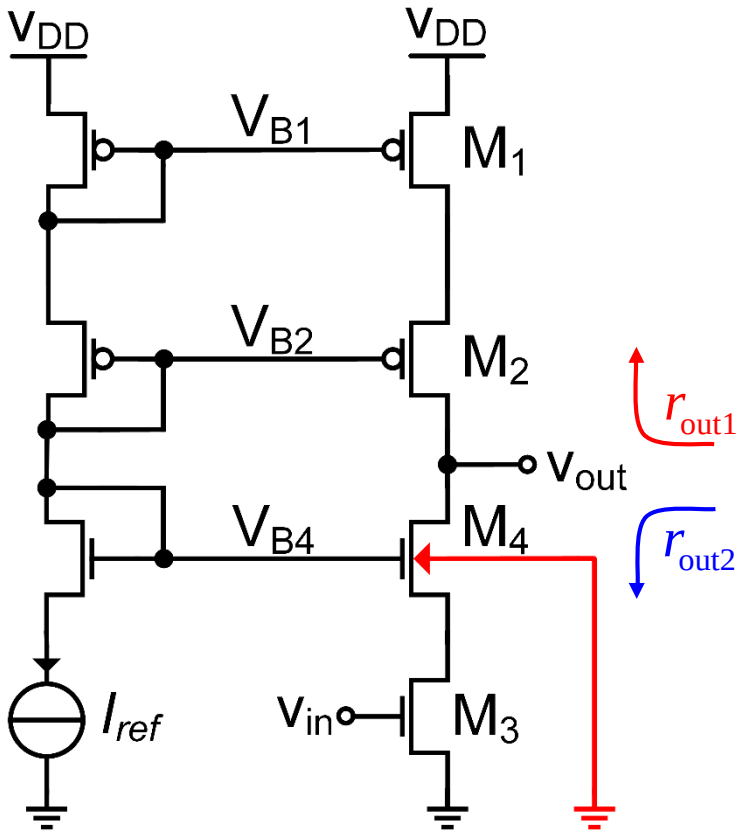
$$\boxed{A_V \approx -g_{m3} r_{ds3}}$$

Increasing r_{out1} only reasonable if r_{out2} is likewise increased!

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Full Cascode CS Amplifier

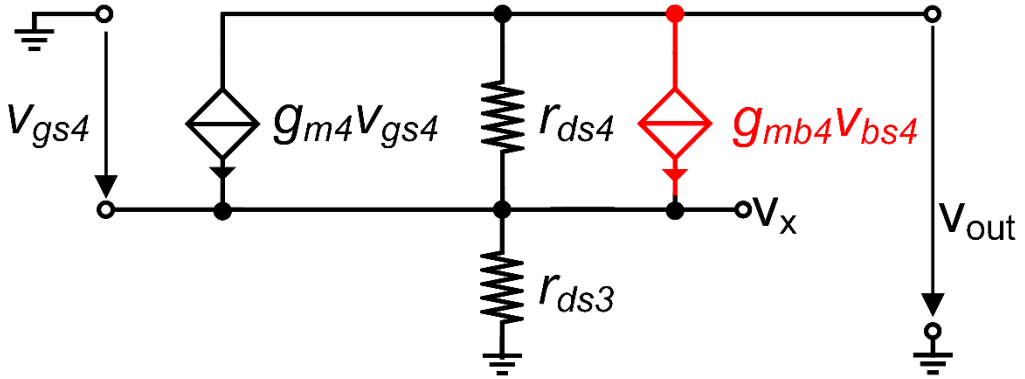


small-signal output resistance

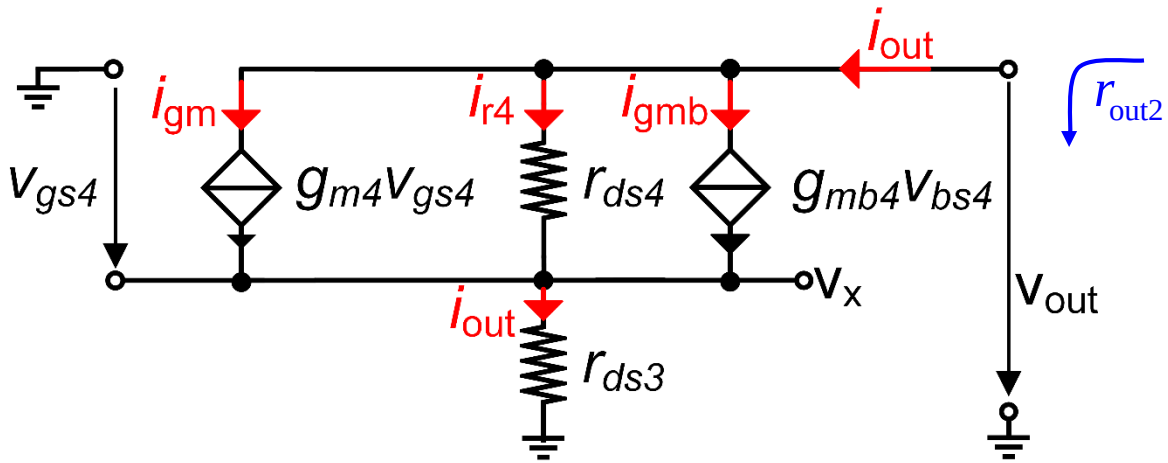
$$r_{\text{out}} = \left.\frac{v_{\text{out}}}{\text{i}_{\text{out}}}\right|_{v_{\text{in}} = 0\ \text{V}} = r_{\text{out1}} || r_{\text{out2}}$$

$$r_{\text{out1}} \approx g_{\text{m2}} r_{\text{ds2}} r_{\text{ds1}}$$

$$r_{\text{out2}} = r_{\text{out1}}?$$



Lower Branch Output Resistance



$$r_{out2} = \left.\frac{v_{out}}{i_{out}}\right|_{v_{in}=0\,V}$$

$$v_{out} = v_x + r_{ds4} i_{r4}$$

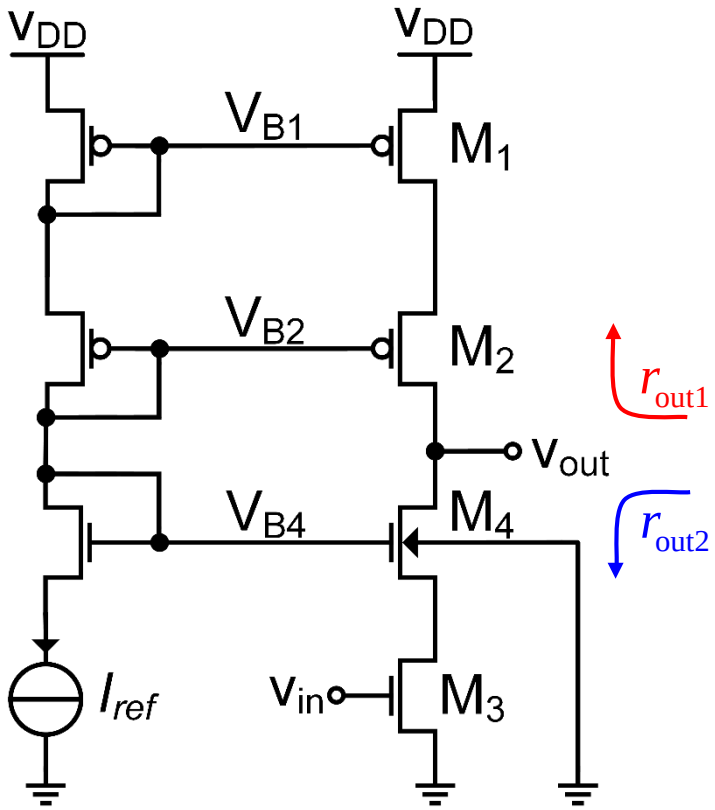
$$v_x = r_{ds3} i_{out} \left(= -v_{gs4} = -v_{bs4}\right)$$

$$\begin{aligned} i_{r4} &= i_{out} - i_{gm} - i_{gmb} \\ &= i_{out} - g_{m4} v_{gs4} - g_{mb4} v_{bs4} \\ &= i_{out} + (g_{m4} + g_{mb4}) v_x \end{aligned}$$

$$\begin{aligned} v_{out} &= r_{ds3} i_{out} + r_{ds4}[i_{out} + (g_{m4} + g_{mb4}) r_{ds3} i_{out}] \\ &= [r_{ds3} + r_{ds4} + (g_{m4} + g_{mb4}) r_{ds4} r_{ds3}] i_{out} \end{aligned}$$

$$\begin{aligned} r_{out2} &= r_{ds3} + r_{ds4} + (g_{m4} + g_{mb4}) r_{ds4} r_{ds3} \\ &\approx (g_{m4} + g_{mb4}) r_{ds4} r_{ds3} \qquad g_{m4} \approx 5 \ldots 10 g_{mb4} \\ &\approx g_{m4} r_{ds4} r_{ds3} \end{aligned}$$

Small-Signal Output Resistance



$$r_{\text{out1}} = r_{\text{ds1}} + r_{\text{ds2}} + g_{\text{m2}} r_{\text{ds2}} r_{\text{ds1}}$$
$$\approx g_{\text{m2}} r_{\text{ds2}} r_{\text{ds1}}$$

$$r_{\text{out2}} = r_{\text{ds3}} + r_{\text{ds4}} + (g_{\text{m4}} + g_{\text{mb4}}) r_{\text{ds4}} r_{\text{ds3}}$$
$$\approx g_{\text{m4}} r_{\text{ds4}} r_{\text{ds3}}$$

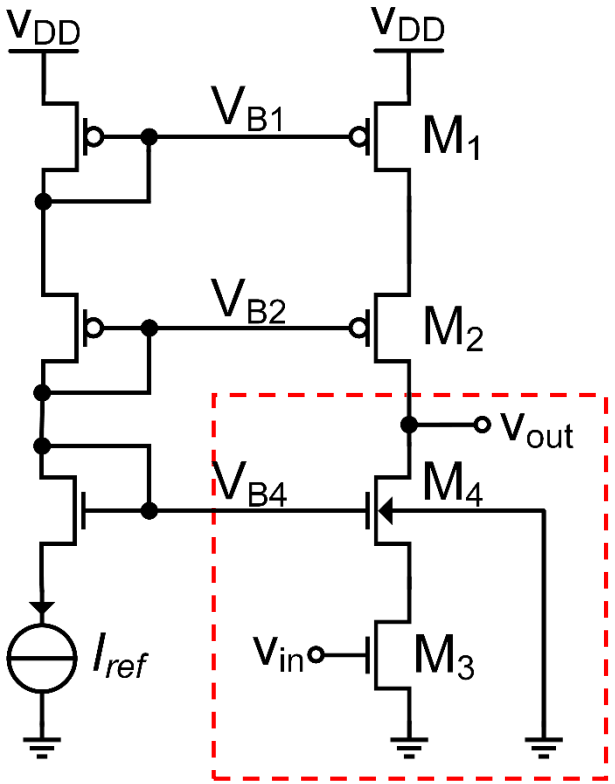
$$r_{\text{out}} \approx g_{\text{m2}} r_{\text{ds2}} r_{\text{ds1}} \parallel g_{\text{m4}} r_{\text{ds4}} r_{\text{ds3}}$$

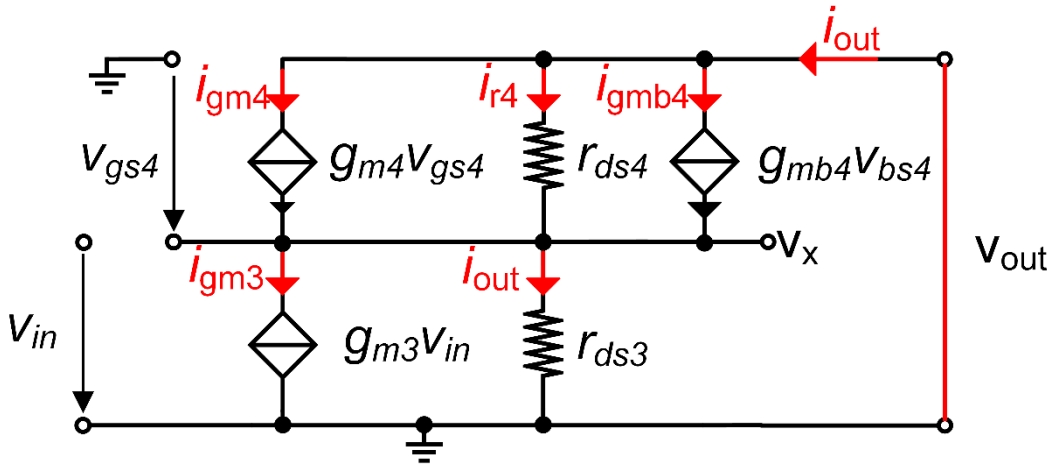
$$\left.\begin{matrix} \text{M}_1 \leftrightarrow \text{M}_3 \\ \text{M}_2 \leftrightarrow \text{M}_4 \end{matrix}\right\} r_{out1} \approx r_{out2}$$

$$\boxed{r_{\text{out}} \approx \frac{g_{\text{m2,4}} r_{\text{ds2,4}} r_{\text{ds1,3}}}{2}}$$

What about the small-signal gain $A_{\text{V}} = -G_{\text{m}} r_{\text{out}}$?
What is the G_{m} of the full cascode CS amplifier?

Transconductance G_m





$$G_m = \left.\frac{i_{out}}{v_{in}}\right|_{v_{out}=0\,V}$$

note:

$$v_{gs4} = v_{bs4} = -v_x$$

$$i_{gm4} + i_{r4} + i_{gmb4} = i_{gm3} + i_{r3}$$

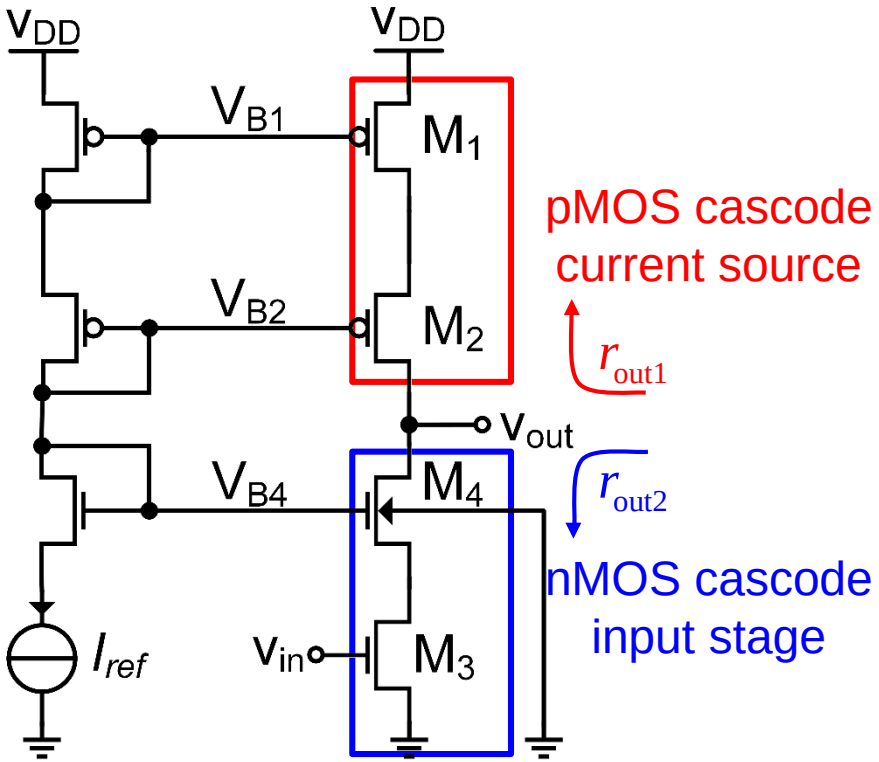
$$\Leftrightarrow -g_{m4}v_x - \frac{v_x}{r_{ds4}} - g_{mb4}v_x = g_{m3}v_{in} + \frac{v_x}{r_{ds3}}$$

$$\Rightarrow v_x = -\frac{g_{m3}}{g_{m4} + g_{mb4} + 1/r_{ds3} + 1/r_{ds4}}\,v_{in}$$

$$G_m \approx g_{m3}$$

$$i_{out} = g_{m3}v_{in} + v_x/r_{ds3} \Rightarrow i_{out} = \left(1 - \frac{1}{1 + r_{ds3}/r_{ds4} + (g_{m4} + g_{mb4})r_{ds3}}\right) g_{m3}v_{in}$$

Small-Signal Amplification



Full cascode CS amplifier

$$r_{\text{out}} \approx g_{\text{m2}} r_{\text{ds2}} r_{\text{ds1}} \parallel g_{\text{m4}} r_{\text{ds4}} r_{\text{ds3}}$$

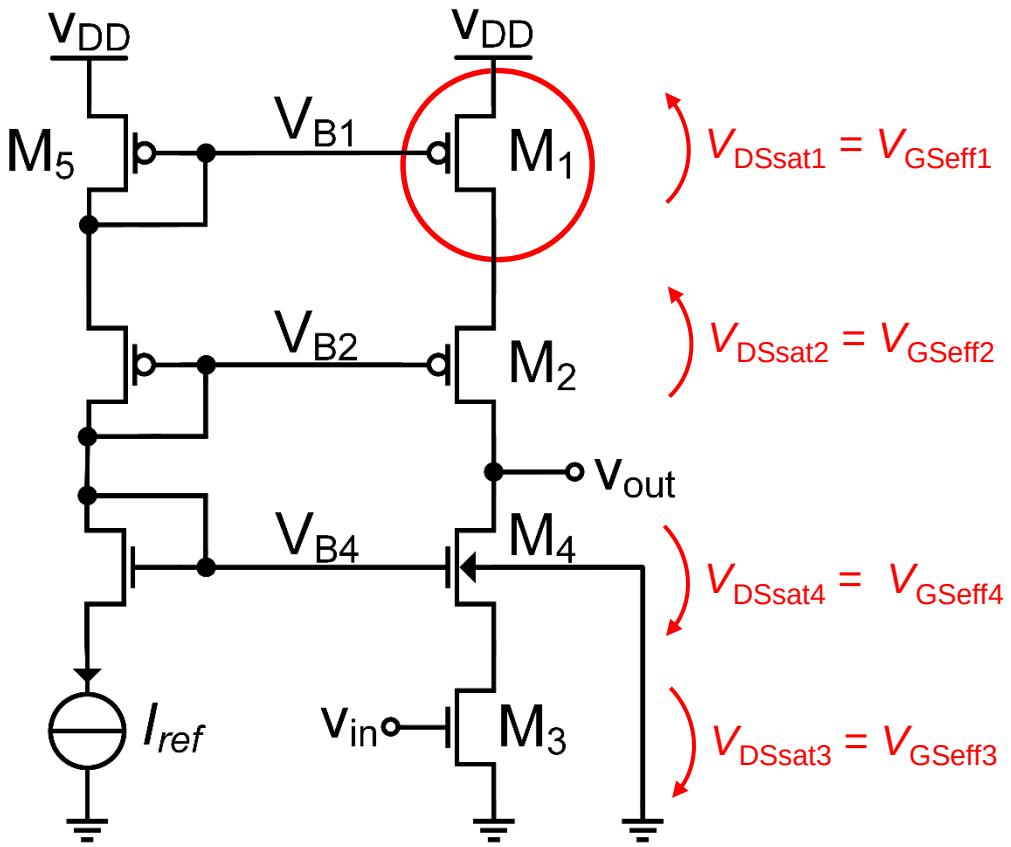
$$\boxed{r_{\text{out}} \approx \frac{g_{\text{m2,4}} r_{\text{ds2,4}} r_{\text{ds1,3}}}{2}}$$

$$\boxed{G_{\text{m}} \approx g_{\text{m3}}}$$

$$\boxed{\begin{aligned} A_{\text{V}} &= -G_{\text{m}} r_{\text{out}} \\ &\approx -g_{\text{m3}} \frac{g_{\text{m2,4}} r_{\text{ds2,4}} r_{\text{ds1,3}}}{2} \end{aligned}}$$

Which transistor defines the bias current?

Ouput Dynamic Range (Signal Swing)



Keep the transistors in saturation!

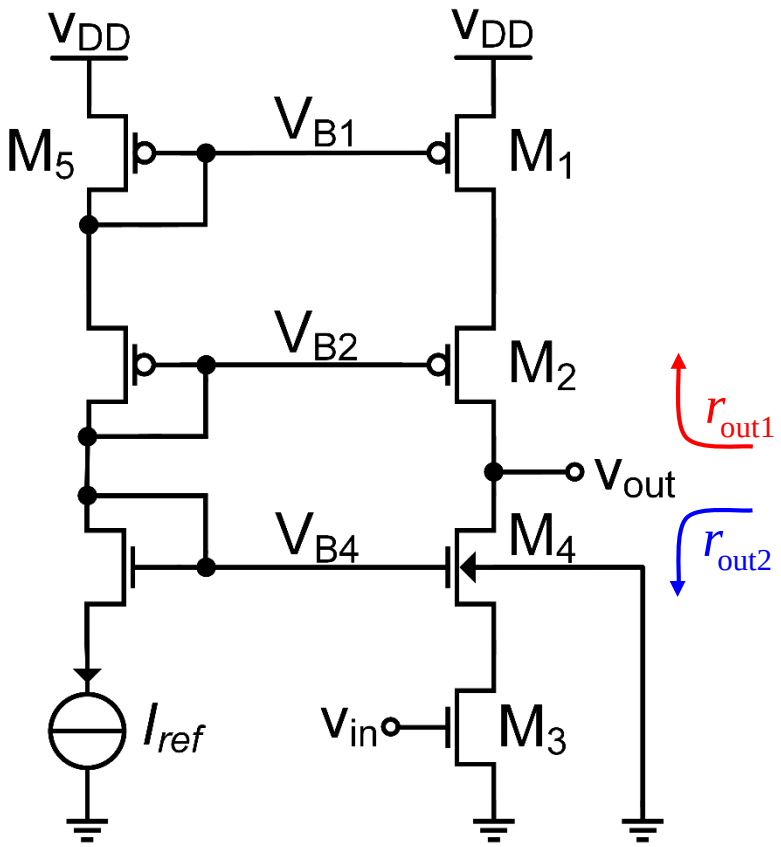
$$2V_{\mathrm{GSeff,n}} < v_{\mathrm{OUT}} < V_{\mathrm{DD}} - 2\left|V_{\mathrm{GSeff,p}}\right|$$

However, $V_{\mathrm{DS1}} = V_{\mathrm{GS1}}$ instead of $V_{\mathrm{DS1}} = V_{\mathrm{GS1}} - V_{\mathrm{th}}$ since we designed the cascode current mirror for $V_{\mathrm{DS1}} = V_{\mathrm{DS5}}$ in order to get a cascode current mirror with a very good matching (s. chapter 3)!

Thus

$$\boxed{2V_{\mathrm{GSeff,n}} < v_{\mathrm{OUT}} < V_{\mathrm{DD}} - 2\left|V_{\mathrm{GSeff,p}}\right| - \left|V_{\mathrm{th,p}}\right|}$$

Increasing the Gain – Further Approaches



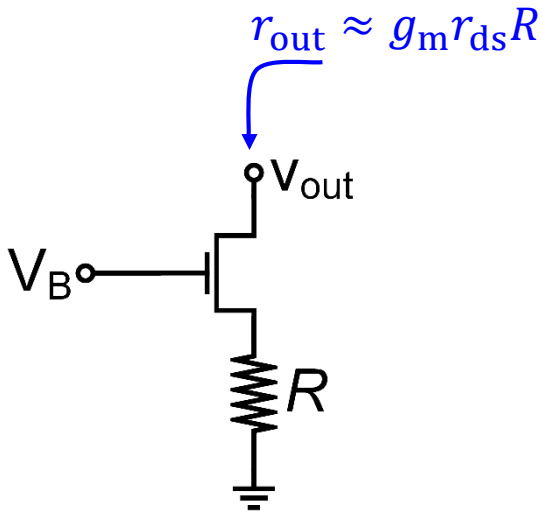
Full cascode CS amplifier

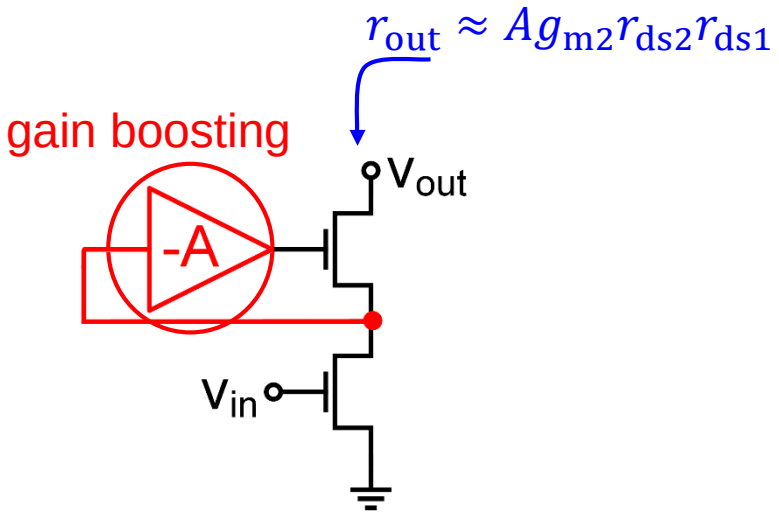
Increase the number of stages = cascading!

- stability

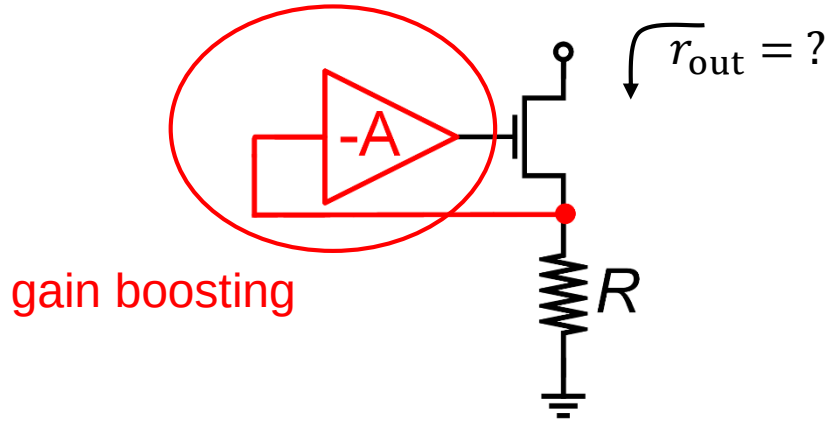
Increase the output resistance!

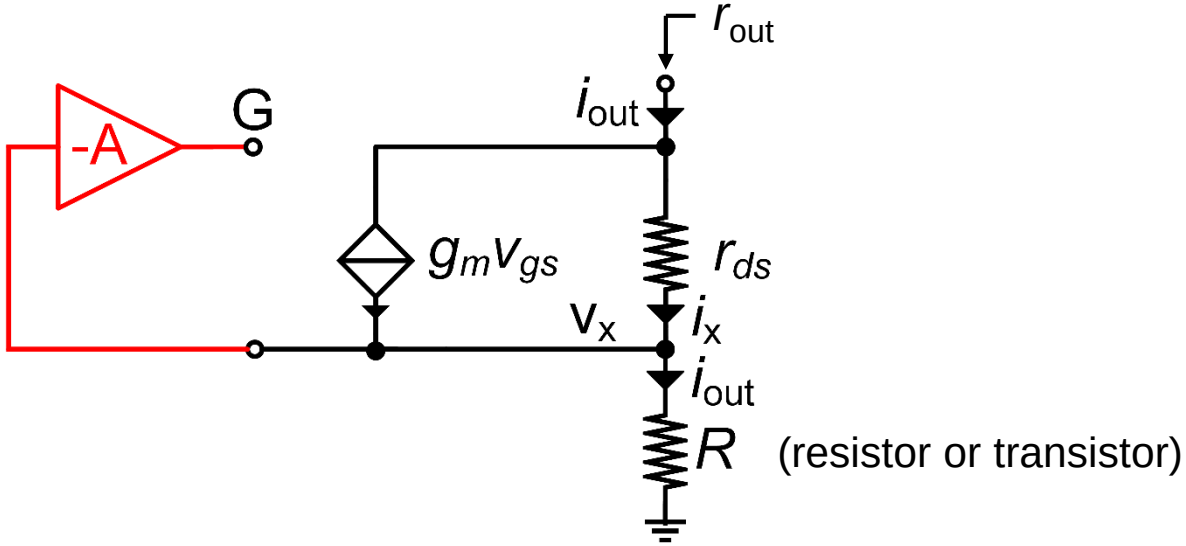
- triple cascode → less output voltage swing
- regulated cascode
  - remember Widlar: source feedback
  - gain boosting: gate feedback





Regulated Cascode – Small-signal Analysis





$$v_{gs} = -Av_x - v_x$$
$$= -(1+A)Ri_{out}$$

$$i_x = i_{out} - g_m v_{gs}$$
$$= [1 + (1+A)\, g_m R] i_{out}$$

$$\frac{v_{out}}{i_{out}} = \frac{r_{ds} i_x + R i_{out}}{i_{out}}$$
$$= [1 + (1+A)\, g_m R] r_{ds}$$

$$\frac{v_{out}}{i_{out}} = r_{out2} \approx A\, g_m r_{ds} R$$

gain boosting — cascode

Summary

- CS amplifier with cascode current source as load element
  - high output impedance, i.e., r_{ds} of the input transistor
  - voltage gain equals the intrinsic gain of a transistor

- full cascode CS amplifier
  - trade-off: very high output impedance ↔ less signal swing

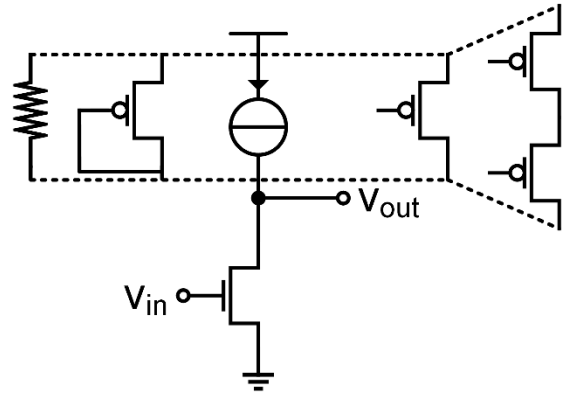
- CS amplifier with gain boosting
  - higher output impedance and thus higher gain by feedback → no further cascode devices

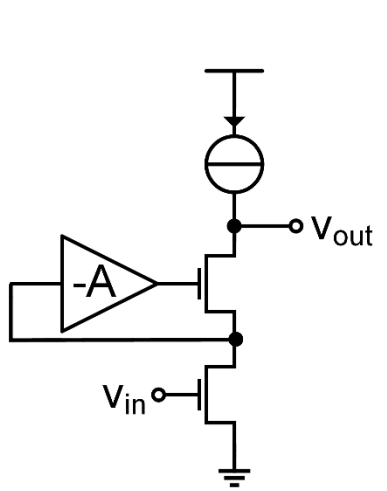
Literature

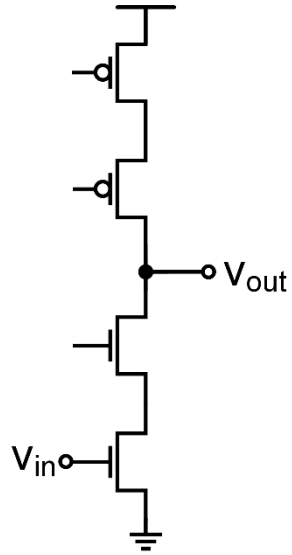
- Razavi, chapters 3.5, page 83-90
- Razavi, chapters 9.4, page 309-313

Common-Source Amplifiers – Summary

Common source stage with …	r_{out}	A_V
resistor load	$= R \| \| r_{ds,in}$	$= -g_m(R \| \| r_{ds,in})$
diode load	$\approx \frac{1}{g_{m,load}}$	$\approx -\sqrt{\frac{\mu_n (W/L)_n}{\mu_p (W/L)_p}}$
ideal current source load	$= r_{ds,in}$	$= -g_{m,in} r_{ds,in}$
transistor current source load	$= r_{ds,in} \| \| r_{ds,load}$	$= \frac{g_{m,in}}{g_{ds,in} + g_{ds,load}}$
cascode current source load	$\approx r_{ds,in}$	$\approx g_{m,in} r_{ds,in}$
full cascode	$\approx \frac{1}{2} r_{ds,in} \underbrace{g_m r_{ds}}_{\text{cascode}}$	$\approx -\frac{(g_{m,in} r_{ds,in})^2}{2}$
gain boosting	$\approx A\, r_{ds,in} \underbrace{g_m r_{ds}}_{\text{cascode}}$	$\approx -A(g_{m,in} r_{ds,in})^2$



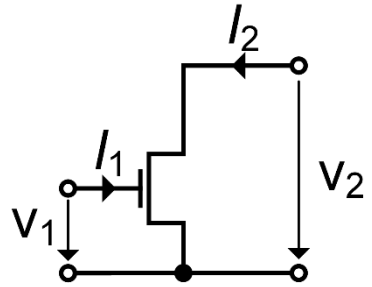




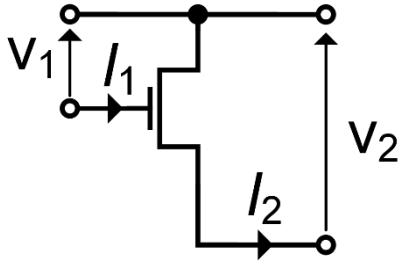
Overview

- Common-Source (CS) Amplifier
- **Common-Drain (CD) Amplifier**
- Common-Gate (CG) Amplifier

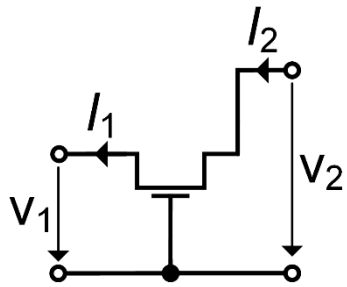
MOSFET – Configurations



common-source configuration (CS)



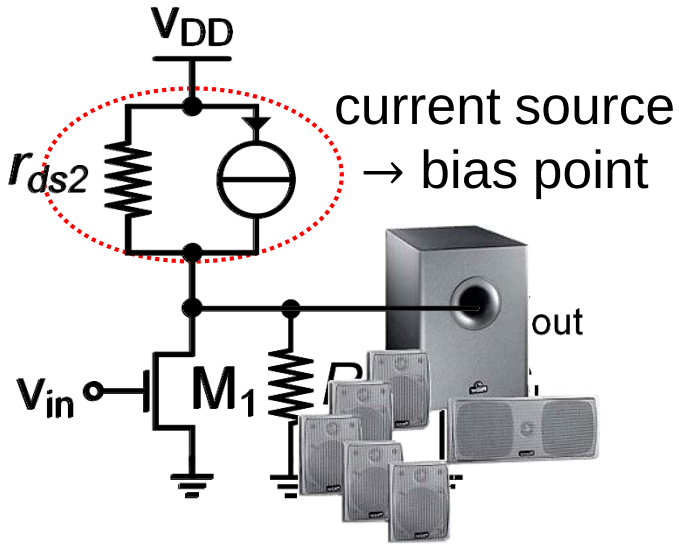
common-drain configuration (CD)



common-gate configuration (CG)

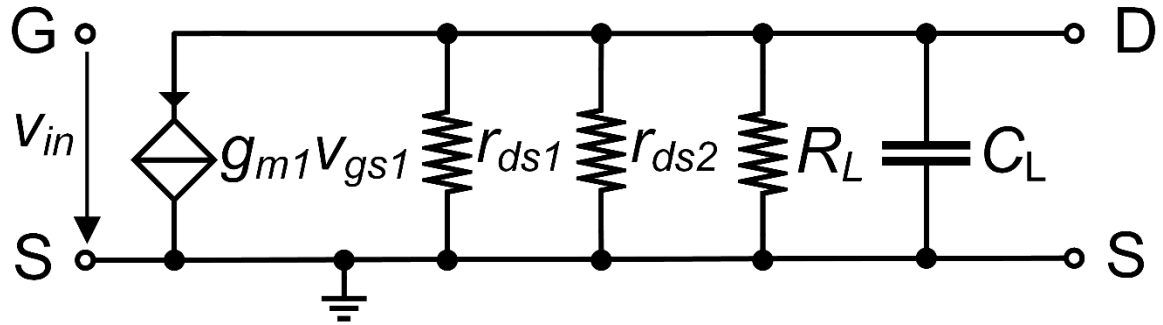
	common-source	common-drain	common-gate
voltage amplification	high	≈ 1	high
current amplification	very high (→ ∞)	very high (→ ∞)	≈ 1
input resistance	very high (→ ∞)	very high (→ ∞)	low
output resistance	average	low	average
phase shift	-180°	0°	0°

Building an Amplifier – Gain-Load Characteristic



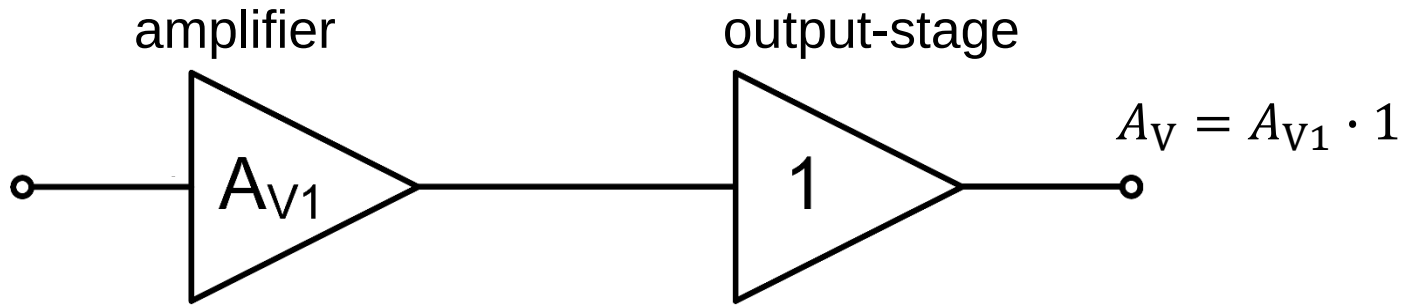
Two usages of the word “load”

- device for defining the operating point
- device to be driven (external load)

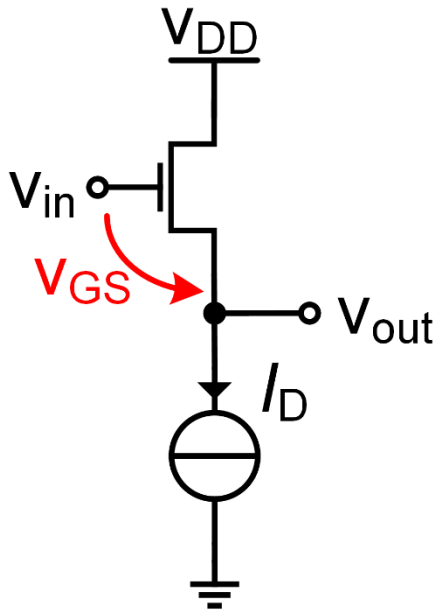


solution: buffer

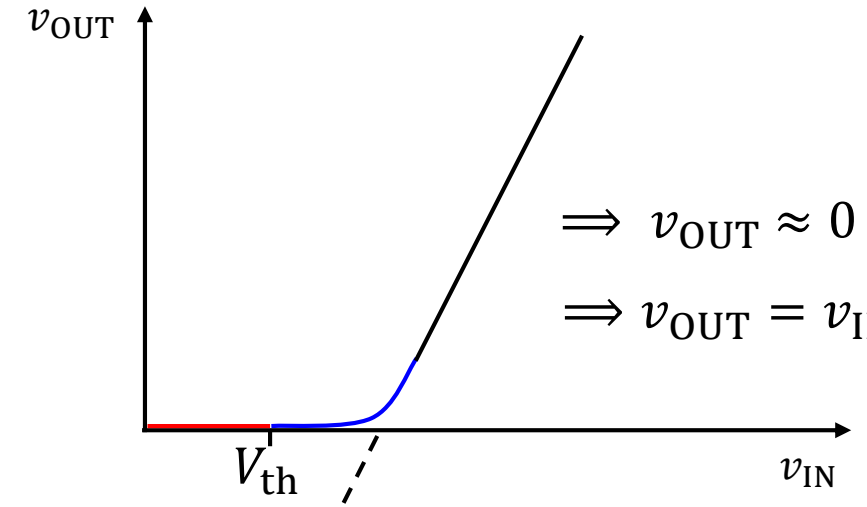
$$A_V = -g_{m1}(r_{ds1} \| r_{ds2} \| R_L) \approx -g_{m1} R_L$$



Large-Signal Behavior – $v_{\mathrm{IN,min}}$



CD amplifier
=
source follower
level shifter



$$\Longrightarrow v_{\mathrm{OUT}} \approx 0\ \mathrm{V}$$

$$\Longrightarrow v_{\mathrm{OUT}} = v_{\mathrm{IN}} \underbrace{-V_{\mathrm{th}} - \sqrt{\frac{2I_{\mathrm{D}}}{\beta_0}\frac{L}{W}}}_{-v_{\mathrm{GS}}}$$

ideal current source &
no supply voltage limitation

$$v_{\mathrm{IN}} < V_{\mathrm{th}} \Longrightarrow i_{\mathrm{D1}} \approx 0\ \mathrm{A}$$

$$v_{\mathrm{IN}} > V_{\mathrm{th}} \Longrightarrow i_{\mathrm{D1}} \approx I_{\mathrm{D}} = \frac{\beta}{2}(v_{\mathrm{GS}} - V_{\mathrm{th}})^2$$

$$= \frac{\beta}{2}(v_{\mathrm{IN}} - v_{\mathrm{OUT}} - V_{\mathrm{th}})^2$$

I_{D} constant $\Longrightarrow v_{\mathrm{GS}} = v_{\mathrm{IN}} - v_{\mathrm{OUT}}$ constant

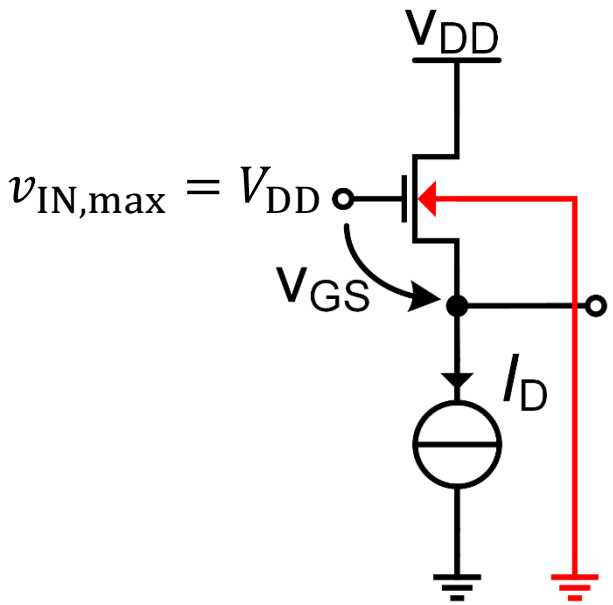
example:

$I_{\mathrm{D}} = 10\ \mu\mathrm{A}$ $\quad \beta_0 = 370\ \frac{\mu\mathrm{A}}{\mathrm{V}^2}$

$V_{\mathrm{th}} = 0.19\ \mathrm{V}$ $\quad \frac{W}{L} = \frac{10\ \mu\mathrm{m}}{10\ \mu\mathrm{m}}$

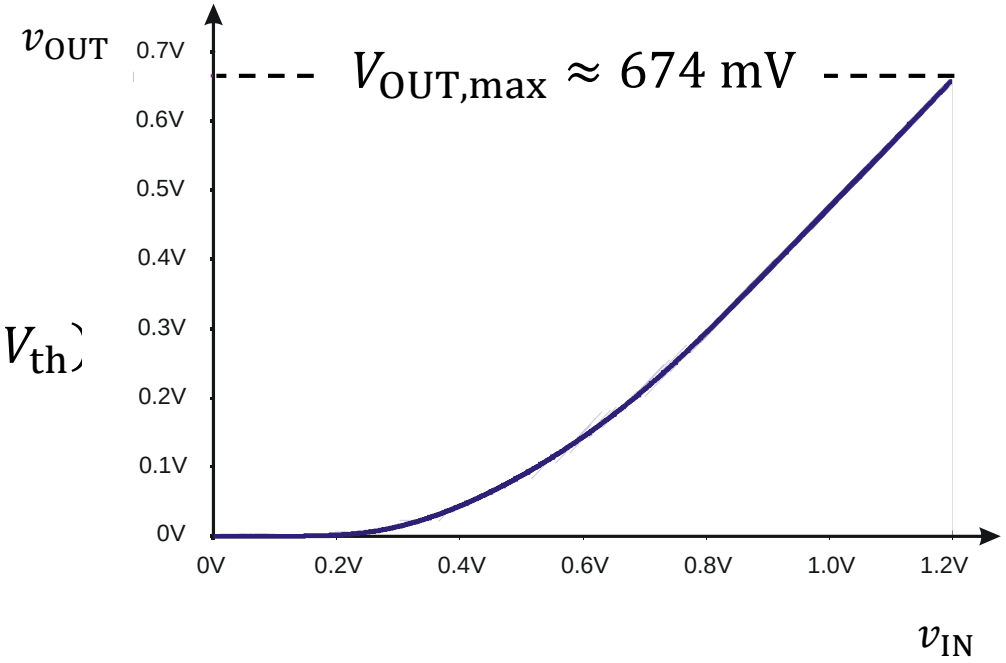
$$\Longrightarrow v_{\mathrm{OUT}} \approx v_{\mathrm{IN}} - 0.42\ \mathrm{V}$$

Large-Signal Behavior – $v_{IN,max}$



$v_{IN,max} = V_{DD}$

$$v_{OUT} = v_{IN} - v_{GS}$$
$$= v_{IN} - (V_{GSeff} + V_{th})$$



MSoFolQ
V_{in}=2±3

$$V_{th} = V_{th0} + \gamma\left(\sqrt{v_{SB} + 2\Phi_F} - \sqrt{2\Phi_F}\right)$$
$$\approx V_{th0} + \gamma\sqrt{v_{SB}}$$

$$v_{OUT} = v_{IN} - V_{GSeff} - V_{th0} - \gamma\sqrt{v_{OUT}}$$

$$V_{OUT,max} = \underbrace{V_{DD}}_{v_{IN}} - V_{GSeff} - V_{th0} - \gamma\sqrt{V_{OUT,max}}$$

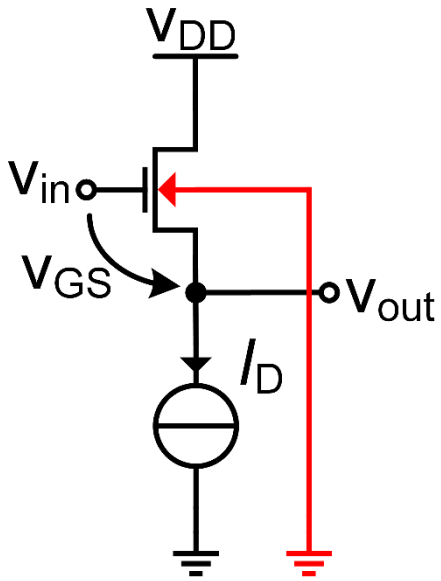
example:

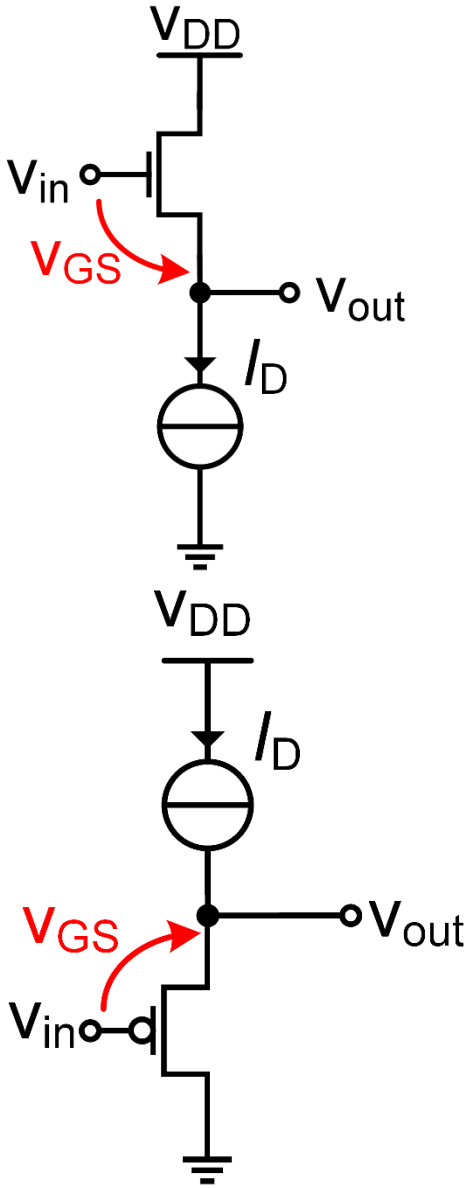
$V_{DD} = 1.2\ \text{V}$ $\quad V_{GSeff} = 0.2\ \text{V}$

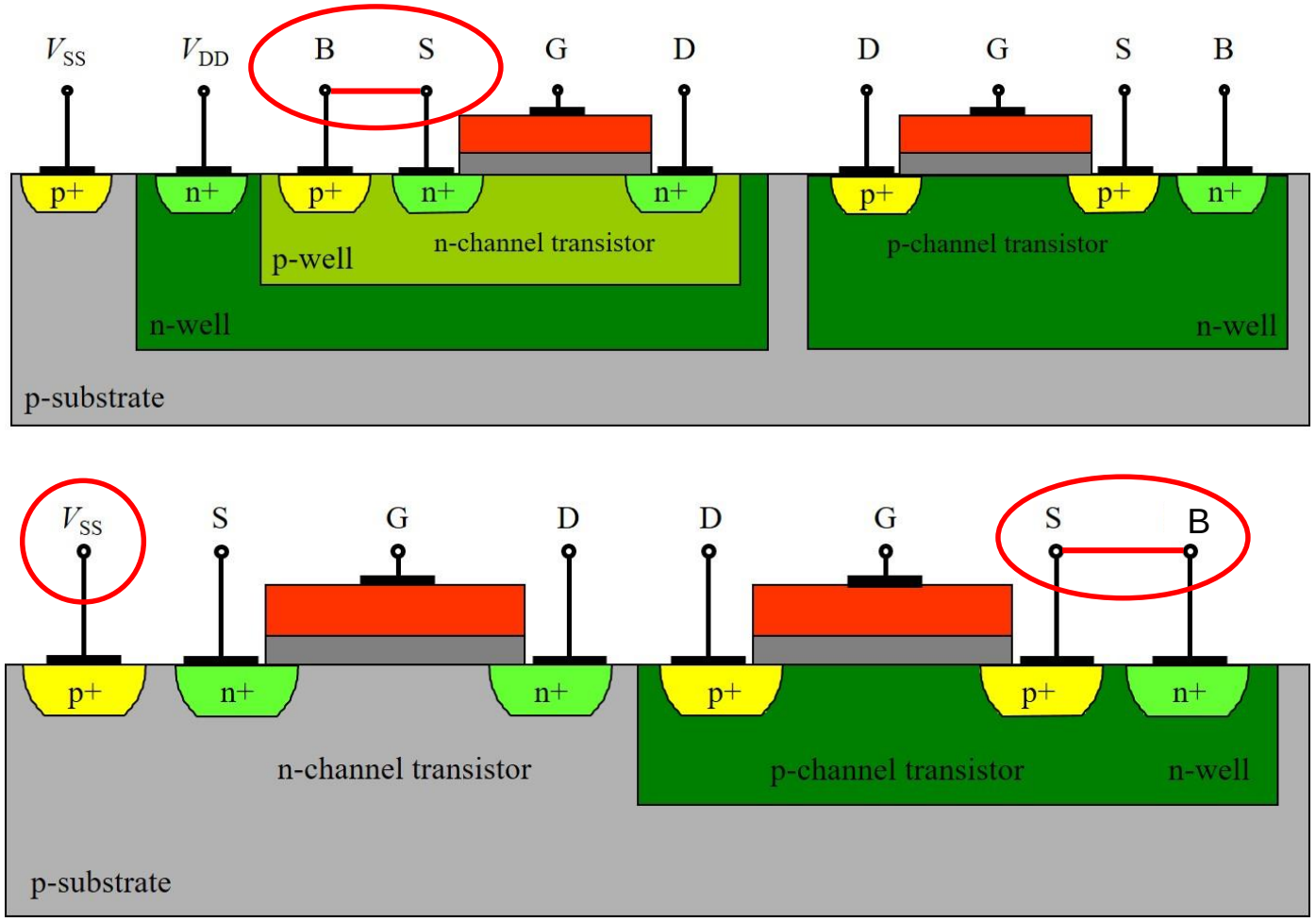
$V_{th0} = 0.19\ \text{V}$ $\quad \gamma = 0.155\ \sqrt{\text{V}}$

$I_D = 10\ \mu\text{A}$

Large-Signal Behavior – $V_{SB} = 0\ \mathrm{V}$

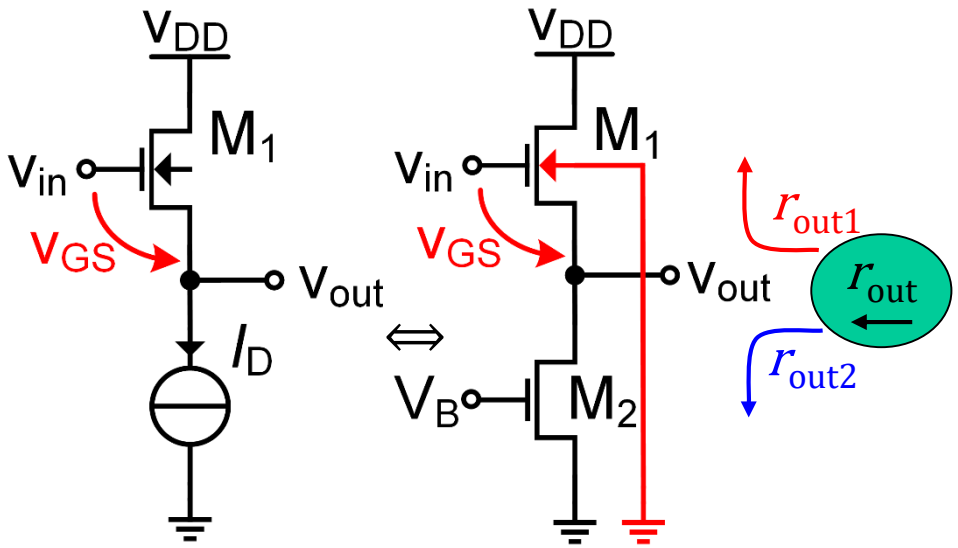


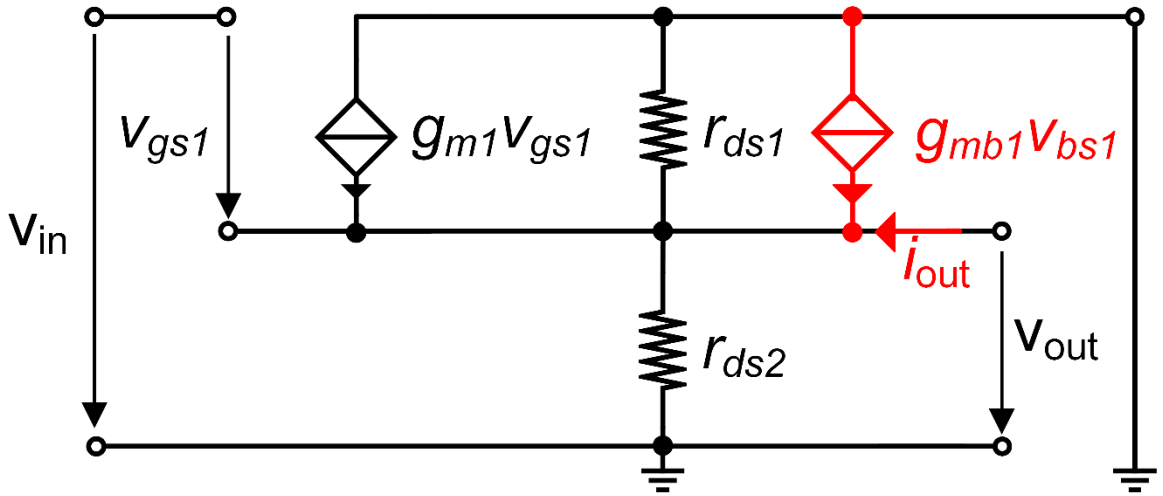




Kuhl, M. (2023). *Chapter 4: Single stage amplifiers* [Lecture slides]. Laboratory of Microelectronics, Department of Microsystems Engineering, IMTEK, University of Freiburg.

CD Amplifier – Small-Signal Behavior





$$A_V = -G_m r_{out}?$$

$$\approx \frac{g_{m1}}{g_{m1}+g_{mb1}} \qquad g_{mb1} \approx 0.1 g_{m1}$$

$$< 1 \approx 0.9$$

$$v_{in} = 0\ \text{V} \rightarrow v_{gs1} = -v_{out} = v_{bs1}$$

$$G_m = \left.\frac{i_{out}}{v_{in}}\right|_{v_{out}=0\ \text{V}} \qquad r_{out} = \left.\frac{v_{out}}{i_{out}}\right|_{v_{in}=0\ \text{V}}$$

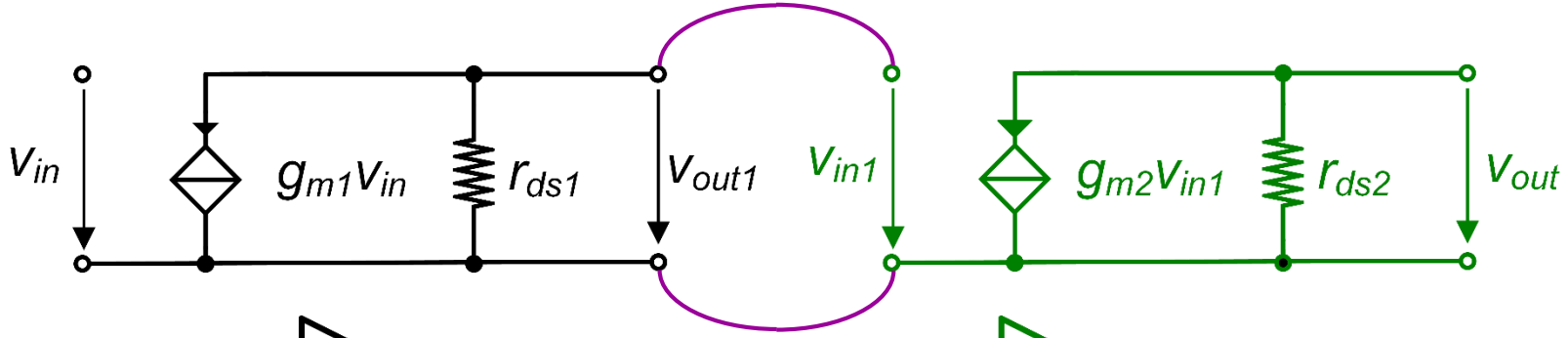
$$i_{out} = -g_{m1} v_{in}$$

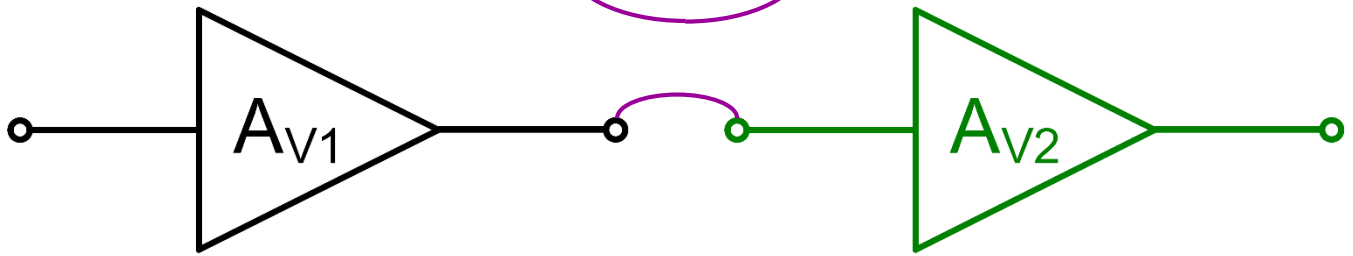
$$r_{out} \approx r_{ds1} \| r_{ds2} \| \frac{1}{g_{m1}} \| \frac{1}{g_{mb1}}$$

$$\boxed{G_m = -g_{m1}}$$

$$\boxed{r_{out} \approx \frac{1}{g_{m1}+g_{mb1}}}$$

Building an Amplifier – Revision





$$A_V = A_{V1}A_{V2} = G_{m1}r_{out1}G_{m2}r_{out2}$$

$A_{V1} = G_{m1}r_{out1}$
amplifier
(high gain, high r_{out})

$A_{V2} = G_{m2}r_{out2}$
buffer
(low gain, low r_{out})

R_L does not affect the gain any longer, yet r_{in2} may!

If $r_{in2} >> r_{out1} \Rightarrow$ no influence from stage 2 on stage 1

Simplification is not allowed if $r_{in2} \leq r_{out1}$!
The small-signal model of the overall system must be analyzed!

Summary

- A CD amplifier is also called a source follower / level-shifter due to its large-signal transfer characteristic.

- A CD amplifier has a high small-signal input and a moderate to low small-signal output resistance → typically used for driving high ohmic loads, i.e., small resistances.
- The signal swing of a CD amplifier is limited!

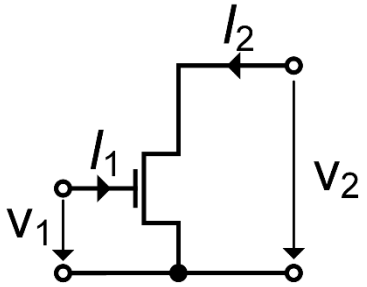
Literature

- Razavi, chapter 3.3, page 67-76

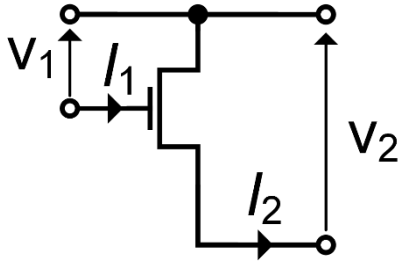
Overview

- Common-Source (CS) Amplifier
- Common-Drain (CD) Amplifier
- **Common-Gate (CG) Amplifier**

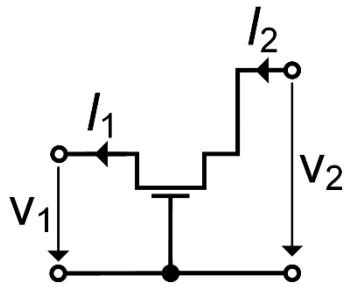
MOSFET – Configurations



common-source configuration (CS)



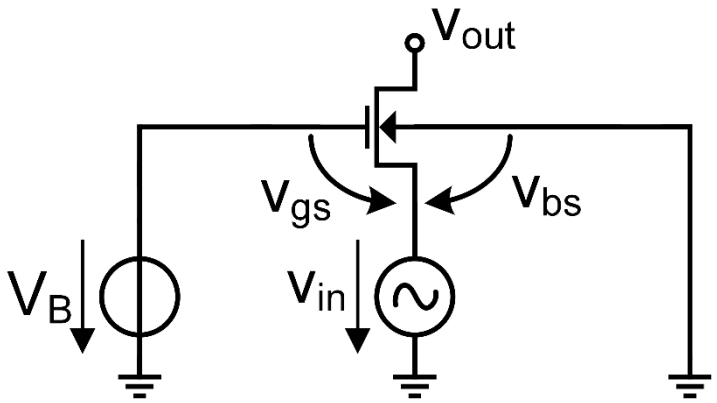
common-drain configuration (CD)



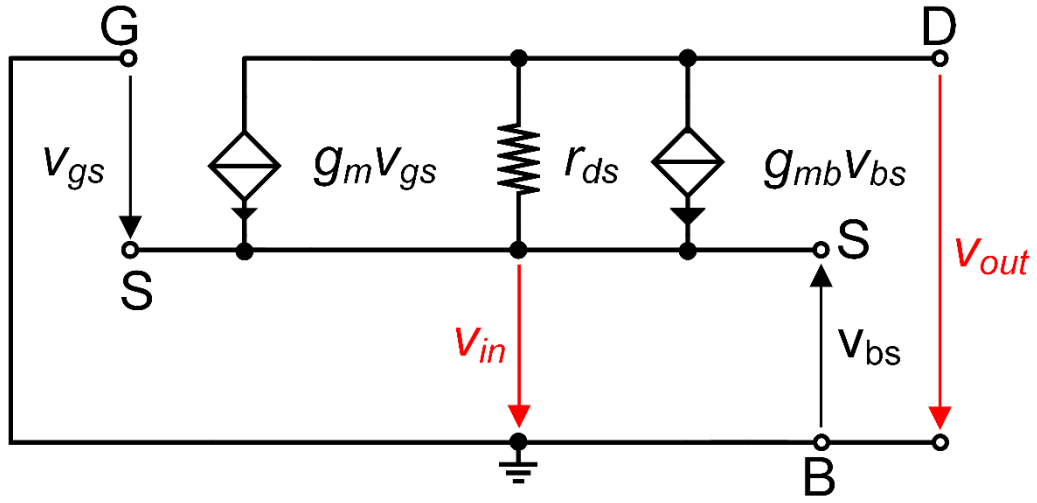
common-gate configuration (CG)

	common-source	common-drain	common-gate
voltage amplification	high	≈ 1	high
current amplification	very high (→ ∞)	very high (→ ∞)	≈ 1
input resistance	very high (→ ∞)	very high (→ ∞)	low
output resistance	average	low	average
phase shift	-180°	0°	0°

Small-Signal Gain



$$A_V = \frac{v_{out}}{v_{in}} = ?$$



$$v_{gs} = v_{bs} = -v_{in}$$

voltage loop:

$$v_{out} = v_{ds} + v_{in}$$

$$-v_{out} - (g_m v_{gs} + g_{mb} v_{bs}) r_{ds} + v_{in} = 0$$

$$\Leftrightarrow -v_{out} + (g_m v_{in} + g_{mb} v_{in}) r_{ds} + v_{in} = 0$$

$$A_V = \frac{v_{out}}{v_{in}} = \cancel{1} + (g_m + \cancel{g_{mb}}) r_{ds} \boxed{\approx g_m r_{ds}}$$

example:

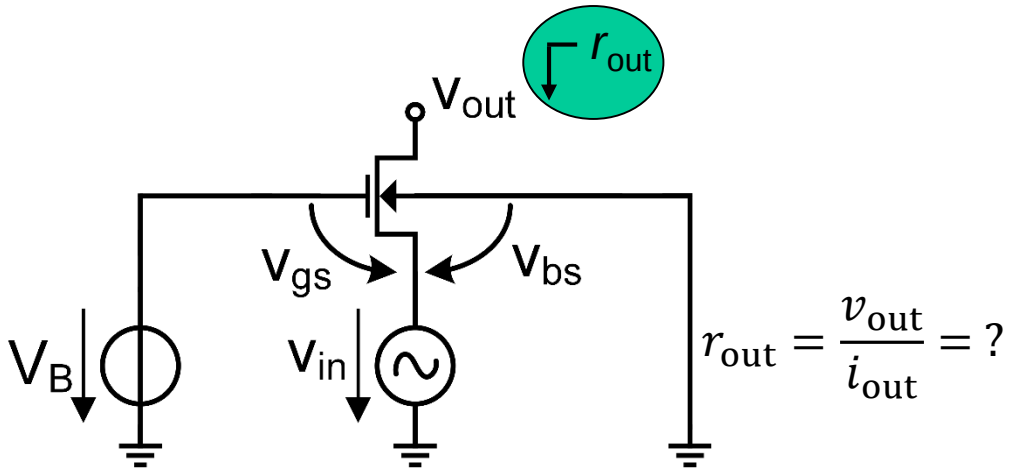
$V_{GS} = 0.35\ \text{V}$ $\quad W/L = 10\mu\text{m}/10\mu\text{m}$

$g_m \approx 58\ \mu\text{S}$ $\quad g_{mb} \approx 8\ \mu\text{S}$

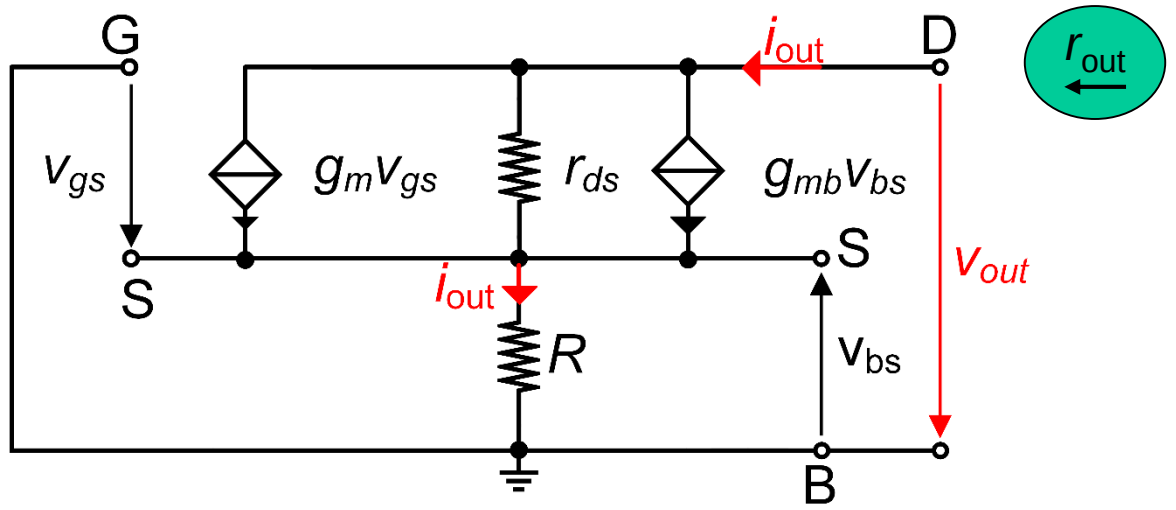
$r_{ds} \approx 8\ \text{M}\Omega$

$A_V \approx g_m r_{ds} = 464$

Output Resistance (@ Drain)



$$r_{out} = \frac{v_{out}}{i_{out}} = ?$$



voltage loop:

$$v_{out} = v_{ds} + v_R$$

$$v_{out} = r_{ds}\left(i_{out} - g_m v_{gs} - g_{mb} v_{bs}\right) + R i_{out}$$

$$\Leftrightarrow v_{out} = r_{ds}(1 + g_m R + g_{mb} R) i_{out} + R i_{out}$$

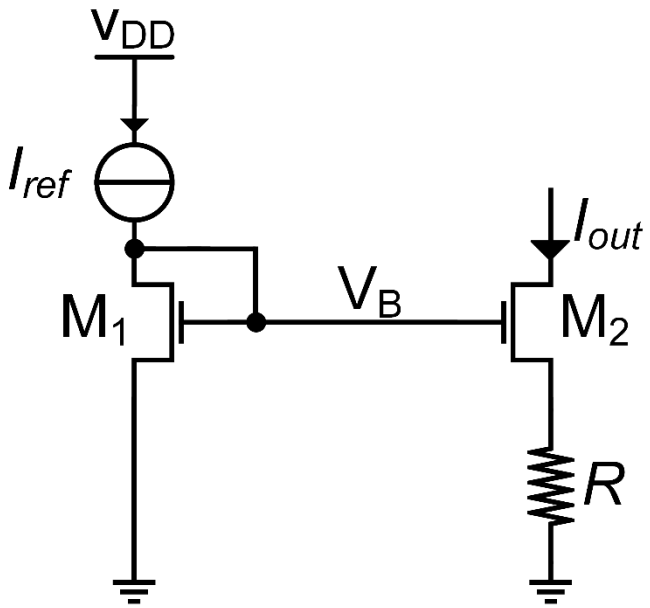
$$r_{out} = \frac{v_{out}}{i_{out}} = [1 + (g_m + g_{mb}) r_{ds}] R + r_{ds}$$

$$\approx g_m r_{ds} R = A_V R$$

$$v_{gs} = v_{bs} = -R i_{out}$$

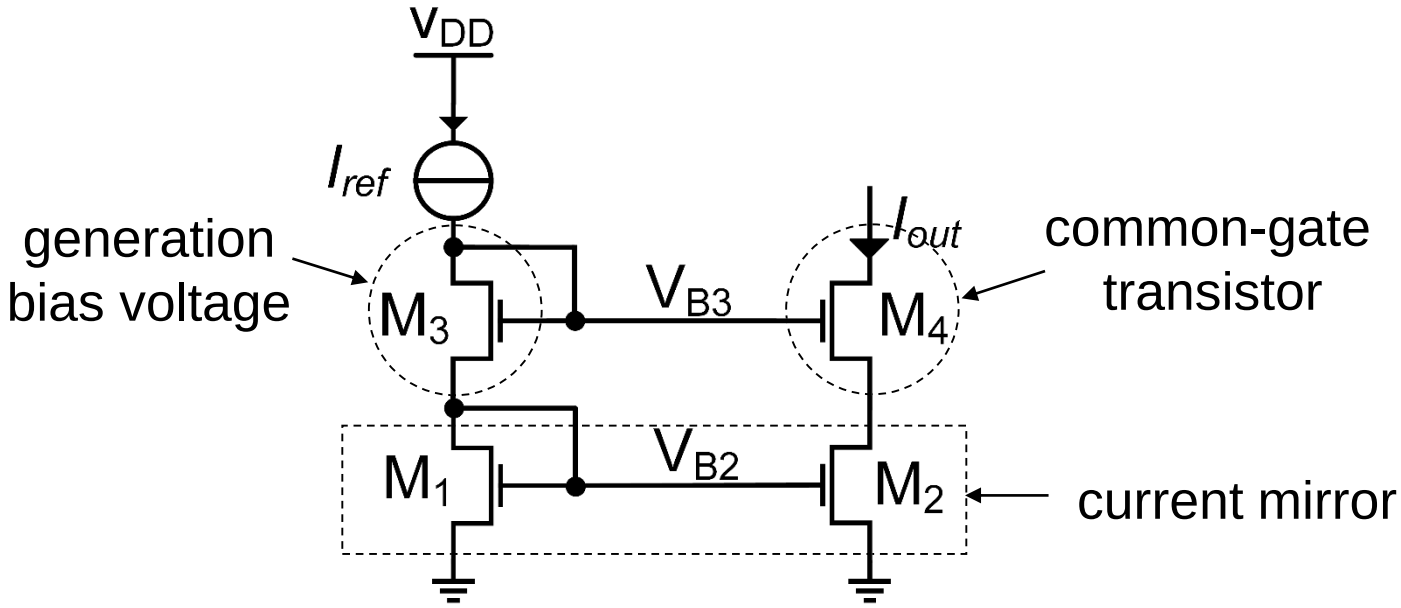
The source resistance is multiplied by the intrinsic gain of the CG amplifier!
--> high to very high output impedance

Application Examples



Widlar current source

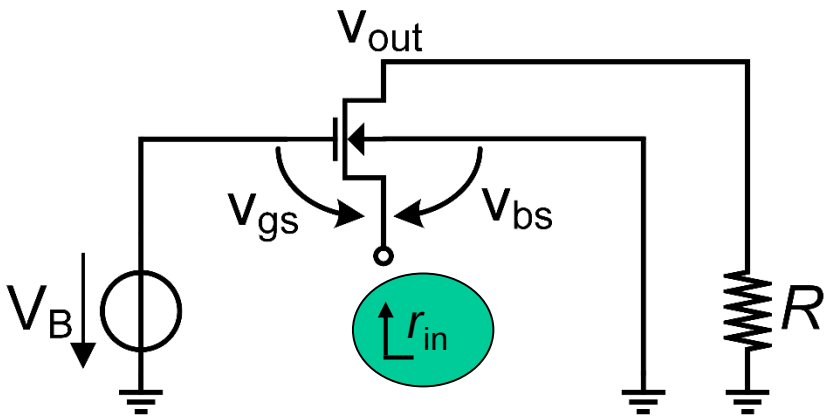
$$r_{\text{out}} = g_{\text{m2}} r_{\text{ds2}} R$$



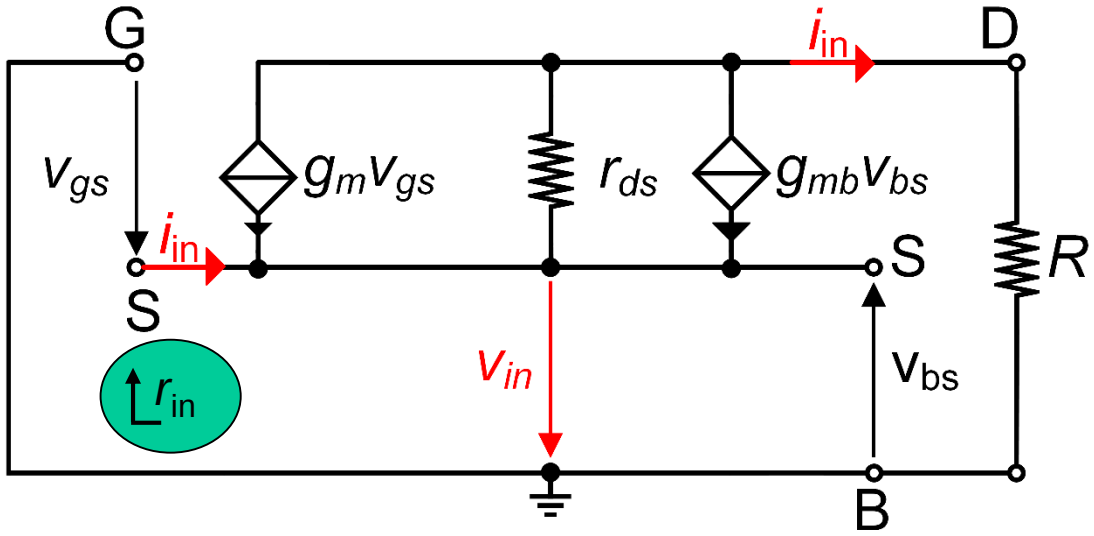
cascode current source

$$r_{\text{out}} = g_{\text{m3}} r_{\text{ds3}} r_{\text{ds2}}$$

Input Resistance (@ Source)



$$r_{\text{in}} = \frac{v_{\text{in}}}{i_{\text{in}}} = ?$$



voltage loop:

$$v_{\text{in}} = v_{\text{R}} - v_{\text{ds}}$$

$$v_{\text{in}} = Ri_{\text{in}} + r_{\text{ds}}\left(i_{\text{in}} + g_{\text{m}}v_{\text{gs}} + g_{\text{mb}}v_{\text{bs}}\right)$$

$$\Leftrightarrow v_{\text{in}} = Ri_{\text{in}} + r_{\text{ds}}(i_{\text{in}} - g_{\text{m}}v_{\text{in}} - g_{\text{mb}}v_{\text{in}})$$

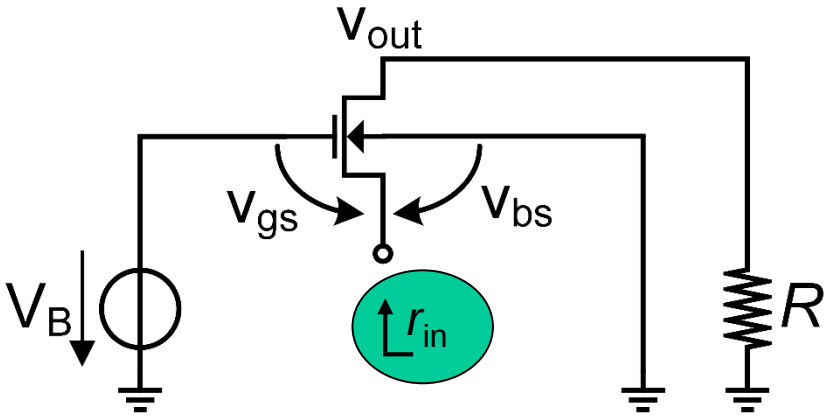
$$r_{\text{in}} = \frac{v_{\text{in}}}{i_{\text{in}}} = \frac{R + r_{\text{ds}}}{\cancel{1} + (g_{\text{m}} + \cancel{g_{\text{mb}}})r_{\text{ds}}}$$

$$\approx \frac{R + r_{\text{ds}}}{g_{\text{m}}r_{\text{ds}}} = \frac{R + r_{\text{ds}}}{A_{\text{V}}}$$

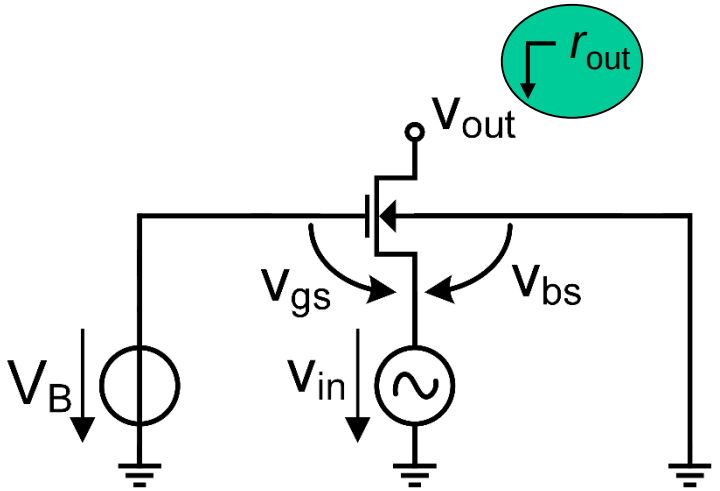
$$v_{\text{gs}} = v_{\text{bs}} = -v_{\text{in}}$$

The overall series resistance $R + r_{\text{ds}}$ is divided by the intrinsic gain of the CG amplifier!
--> low to very low input impedance

Current Buffer



$$r_{\text{in}} = \frac{v_{\text{in}}}{i_{\text{in}}} \approx \frac{R + r_{\text{ds}}}{g_{\text{m}} r_{\text{ds}}} = \frac{R + r_{\text{ds}}}{A_{\text{V}}}$$



$$r_{\text{out}} = \frac{v_{\text{out}}}{i_{\text{out}}} \approx g_{\text{m}} r_{\text{ds}} R = A_{\text{V}} R$$

$i_{\text{out}} = i_{\text{in}}$

small r_{in}

large r_{out}

$\Longrightarrow$ **current buffer**

Summary

- equivalent resistance at drain $r_{\mathrm{out}} \approx A_{\mathrm{V}} R$
- equivalent resistance at drain $r_{\mathrm{in}} \approx \frac{R + r_{\mathrm{ds}}}{A_{\mathrm{V}}}$
- typical application: current buffer

Literature

- Razavi, chapter 3.4, page 76-83